

Analysis I: One Variable

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0 Writing a Proof

In order to do mathematics, it is essential being able to write a proof. In this section we will discuss some basic rules and guidelines for writing proofs. The chapter is based on [2, 3, 4].

After having written a proof, typically it is in a state of exploration. This is the first draft of the proof. The next step is to write down a clean version of the proof. The latter of which we will now focus on.

The most important rule is probably to keep in mind normal grammar. Write in full sentences, use punctuation and capital letters. A proof is a piece of literature. It should be readable and understandable.

Definition 0.1: Definition

A **DEFINITION** is a precise description of a mathematical object.

A mathematical definition should be precise and unambiguous. It is used to abbreviate long descriptions. For example, we can define prime numbers to avoid constantly repeating: “Positive integers $p > 1$ which only have 1 and p as divisors”. Simply said, definitions help us talk about and understand mathematical proofs.

Definition 0.2: Theorem, Lemma, Proposition and Corollary

A **THEOREM** is a proven mathematical statement.

A **LEMMA** is a theorem which is used to prove another theorem.

A **PROPOSITION** is a theorem of lesser importance.

A **COROLLARY** is a theorem which follows directly from another theorem.

Theorems, lemmas, propositions and corollaries are all statements which have been proven to be true. The difference between them is their relevance. This already implies that the names are somewhat subjective. Together with definitions, they provide the building blocks of a proof.

Two words that require extra attention are **SUFFICIENT** and **NECESSARY**. If we have $A \Rightarrow B$ then A is a sufficient for B and B is a necessary condition for A . It’s best to simply memorize this.

0.1 Labelling

It is common to label certain objects, for example a line l or a matrix A . This is done to avoid repeating long descriptions however it is important to follow some rules

when labelling objects. For example, one shouldn’t call a function 2 or a set $9[xP :)$. Instead, one should try to use meaningful labels. This can be done by labelling similar objects in a similar fashion. Similarly, one should use standards when they exist. For example, it is common to use $\mathbb{N}$ for the set of natural numbers, $\mathbb{Z}$ for the set of integers and so on.

Additionally, it can be useful to implement a hierarchy of labels. For example, one can use capital letters for sets and lower case letters for its elements.

Lastly, it is good practice to not use unnecessary labels. For example, one doesn’t need to write “Every differentiable function f is continuous.” Instead, one can simply write “Every differentiable function is continuous.”

0.2 Proofing Techniques

A proof is a logical argument which shows that a certain statement is true. There are many possible techniques to do so, the most common ones being listed below.

Definition 0.3: Direct Proof

A **DIRECT PROOF** is a proof which starts from the assumptions and uses logical steps to arrive at the conclusion.

A direct proof is the most straightforward way to prove a statement.

Example 0.4:

We want to show that the sum of two even numbers is even. Let a, b be two even numbers. By definition of even numbers, there exists $k, m \in \mathbb{Z}$ such that $a = 2k$ and $b = 2m$. Thus,

$$a + b = 2k + 2m = 2(k + m).$$

As $k + m \in \mathbb{Z}$, we have shown that $a + b$ is even.

If one has to proof a statement for the set of all natural numbers, or similar sets, one can use induction.

Definition 0.5: Proof by Induction

A **PROOF BY INDUCTION** is a proof which consists of two steps.

- 1: Check it is true for $n = 1$.
- 2: If the equation holds for some n , then it holds for $n + 1$.

This technique is extremely powerful as can be seen in the following example.

Example 0.6: Gaussian Counting

We want to show that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$.

Base Case: Let $n = 1$. Then, $1 = \frac{1(1+1)}{2} = 1$. Thus, the equation holds for $n = 1$.

Induction Step: Fix $n \geq 1$. We know that $1 + 2 + \dots + n = \frac{n(n+1)}{2}$. We want to show that $1 + 2 + \dots + n + (n+1) = \frac{(n+1)(n+2)}{2}$.

Notice that the first block of the LHS is exactly what we have assumed in the induction hypothesis. Thus, we can replace it.

$$\begin{aligned} \frac{n(n+1)}{2} + (n+1) &= \frac{n^2 + n}{2} + n + 1 \\ &= \frac{n^2 + n + 2n + 2}{2} \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

Thus, the right hand side equals the left hand side.

Definition 0.7: Proof by Contradiction

A **PROOF BY CONTRADICTION** is a proof which starts by assuming the negation of the statement to be proven and then derives a contradiction.

Classic examples of a proof by contradiction are that $\sqrt{2}$ is irrational or that there are infinitely many prime numbers.

Example 0.8: $\sqrt{2}$ is irrational

We want to show that $\sqrt{2}$ is irrational. We do so by assuming the opposite, i.e. that $\sqrt{2}$ is rational. Thus, we can write $\sqrt{2} = \frac{p}{q}$ where $p, q \in \mathbb{Z}$ have no common divisors. Squaring both sides, we get $2q^2 = p^2$. This implies that p^2 is even, which in turn implies that p is even.

Thus, we can write $p = 2k$ for some $k \in \mathbb{Z}$. Plugging this back into the equation, we get $2q^2 = 4k^2$, or $q^2 = 2k^2$. This implies that q is even. However, this contradicts our initial assumption that p and q have no common divisors. ζ

Thus, $\sqrt{2}$ is irrational.

Example 0.9: Infinitely many primes

We want to show that there are infinitely many prime numbers. Let us assume the opposite, i.e. that there are finitely many prime numbers.

Let $M = \{p_1, p_2, \dots, p_n\}$ be the set of all prime numbers. Further, let $k := p_1 p_2 \dots p_n$. Thus, every prime number divides k . However, consider $k + 1$. This number is not divisible by any prime number, as it always leaves a remainder of 1. Thus, $k + 1$ is either prime itself or has a prime divisor which itself is not in M . ζ

Thus, there are infinitely many prime numbers.

1 Introduction

We begin our journey by an exercise.

Example 1.1:

Consider the x, y plane and the parabola given by $y = x^2$. Consider the region bounded by the parabola and the x -axis and the vertical lines $x = 0$ and $x = 1$. i.e. the region

$$P = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq x^2\}.$$

To answer this question, we need to know what area is. For this we define three principles which we want the area to satisfy. Whatever it is, it should satisfy:

- $[a, b] \times [c, d]$ has area $(b - a)(d - c)$,
- if F is contained in G , then $\text{Area}(F) \leq \text{Area}(G)$,
- if F and G are disjoint, then the $\text{Area}(F \cup G) = \text{Area}(F) + \text{Area}(G)$.

The set P in the plane

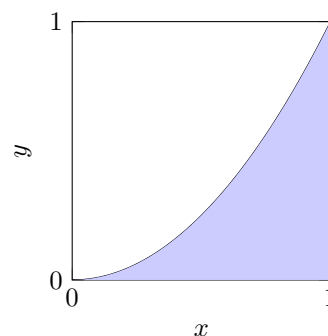


Figure 1: The set P in the plane.

Using this, we can now answer the question.

Example 1.1:

With these three principles, the area of P is $\frac{1}{3}$.

To prove this, we will need to have to use the following lemma.

Lemma 1.2:

For $n \geq 1$,

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}.$$

We want to use **INDUCTION**, which is a proof that is split into two steps.

Proof.

Base Case: Let $n = 1$.

$$\begin{aligned} LHS &= 1^2 = 1 \\ RHS &= \frac{1}{3} + \frac{1}{2} + \frac{1}{6} = 1 \end{aligned}$$

Induction Step: Fix $n \geq 1$. We know that $1^2 + 2^2 + \dots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$. We want $1^2 + 2^2 + \dots + n^2 + (n+1)^2 = \frac{(n+1)^3}{3} + \frac{(n+1)^2}{2} + \frac{n+1}{6}$.

We notice, that the first block of the LHS is exactly what we have assumed in the induction hypothesis. Thus, we can replace it.

$$\begin{aligned} \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} + (n+1)^2 &= \frac{(n+1)^3}{3} + \frac{(n+1)^2}{2} + \frac{n+1}{6} \\ LHS &= \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} + n^2 + 2n + 1 \\ &= \frac{n^3}{3} + \frac{3n^2}{2} + \frac{13n}{6} + 1 \\ RHS &= \frac{n^3 + 3n^2 + 3n + 1}{3} + \frac{n^2 + 2n + 1}{2} \\ &\quad + \frac{n+1}{6} \\ &= \frac{n^3}{3} + \frac{3n^2}{2} + \frac{13n}{6} + 1 \end{aligned}$$

Thus, the right hand side equals the left hand side. $\square$

Using this lemma, we now want to calculate the area of our parabola.

Proof. [Proof of Theorem 1.1] To calculate the area of P we will fix a $n \geq 1$ and split the interval $[0, 1]$ into n equal parts.

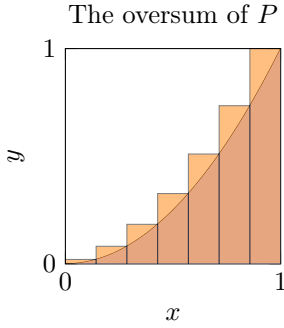


Figure 2: The oversum of P .

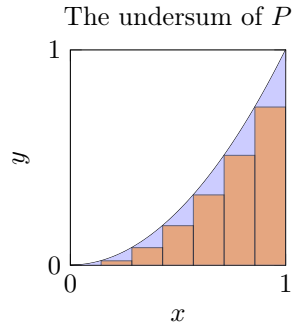


Figure 3: The undersum of P .

Let $\mathcal{A} = \text{Area}(P)$. We have

$$\mathcal{A} \leq \text{Area}(\text{figure 2}).$$

$$\begin{aligned} \mathcal{A} &\leq \frac{1}{n} \frac{1}{n^2} + \frac{1}{n} \frac{2^2}{n^2} + \dots + \frac{1}{n} \frac{n^2}{n^2} \\ &= \frac{1}{n^3} (1^2 + 2^2 + \dots + n^2) \\ &= \frac{1}{n^3} \left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} \right) \\ &= \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \end{aligned}$$

Similarly, for the second figure,

$$\begin{aligned} \mathcal{A} &\geq \text{Area}(\text{figure 3}) \\ &= \frac{1}{n} \left(\left(\frac{0}{n} \right)^2 + \left(\frac{1}{n} \right)^2 + \dots + \left(\frac{n-1}{n} \right)^2 \right) \\ &= \frac{1}{n^3} (0^2 + 1^2 + \dots + (n-1)^2) \\ &= \frac{1}{n^3} \left(\frac{(n-1)^3}{3} + \frac{(n-1)^2}{2} + \frac{n-1}{6} \right) \\ &= \frac{1}{n^3} \left(\frac{n^3}{3} - n^2 + \frac{n}{6} \right) \\ &= \frac{1}{3} - \frac{1}{n} + \frac{1}{6n^2} \end{aligned}$$

Where in the last step we used the lemma with $n-1$ instead of n . Combining the both formulas, we have

$$-\frac{1}{2n} + \frac{1}{6n^2} \leq \mathcal{A} - \frac{1}{3} \leq \frac{1}{2n} + \frac{1}{6n^2}.$$

The only reasonable number between these two bounds is $\mathcal{A} = \frac{1}{3}$. $\square$

For completeness, we give a definition of a set in naive set theory.

Definition 1.3: Set

A **SET**

- consists of distinct **ELEMENTS**,
- is uniquely characterized by its elements,
- is not an element of itself.
- If $A(x)$ is a statement about elements x in a set X , then $\{x \in X \mid A(x)\}$, denotes all elements for which $A(x)$ is true.

Finally, $\emptyset$ is the **EMPTY SET**.

The most useful property is probably the last one. An example of a set is $\{1, 2, 7, 10\}$, or $X = \{n \in \mathbb{N} \mid n \text{ is even}\} = \{n \in \mathbb{N} \mid \exists m \in \mathbb{N} : n = 2m\}$.

2 The Real Numbers

Lec 2

2.1 The Axioms of the Real Numbers

We want to study properties of the real numbers to formally describe them. To do this we introduce the Group.

Definition 2.1: Group

A **GROUP** is a set G with one operation $*$ that satisfies:

- Associativity: $(a*b)*c = a*(b*c)$
- Neutral element: $\exists e \in G : a*e = e*a = a \forall a$
- Inverse element: $\forall a \in G \exists a^{-1} \in G : a*a^{-1} = e$

Remark 2.2:

In general $a*b \neq b*a$. If $a*b = b*a \forall a, b \in G$ then G is a **COMMUTATIVE** or **ABELIAN** group.

Some examples of groups:

- $\mathbb{N}$ with addition $+$

We can check this: Addition is associative, the neutral element is 0. However, the inverse of for example 3 is $-3 \notin \mathbb{N}$. So $\mathbb{N}$ is not a group.

- $\mathbb{Z}$ with addition $+$

The first two points still hold. And $(-n) + n = 0$. Therefore $(\mathbb{Z}, +)$ is a commutative group as $n+m = m+n$.

- $\mathbb{Q}^* = \{\frac{p}{q} : p, q \in \mathbb{Z} \setminus \{0\}\}$ with multiplication $\cdot$

This is associative, the neutral element is 1 as $\frac{p}{q} \cdot 1 = 1 \cdot \frac{p}{q} = \frac{p}{q}$. And $\frac{p}{q} \cdot \frac{q}{p} = 1$. Once again this is also commutative. Thus $(\mathbb{Q}^*, \cdot)$ is a commutative group.

Lemma 2.3:

Neutral and inverse are unique.

Proof. Let $e, e' \in G$ be two neutral elements.

$$e' \text{ (as } e \text{ is neutral)} = e * e' = e \text{ (as } e' \text{ is neutral)}.$$

Given $a \in G$, assume b, c are both inverse elements of a . $a*b = b*a = e$ and $a*c = c*a = e$. Then

$$b = b * e = b * (a * c) = (b * a) * c = e * c = c.$$

□

Remark 2.4:

Since $a * a^{-1} = a^{-1} * a = e$, then a is the inverse of a^{-1} . Therefore,

$$(a^{-1})^{-1} = a.$$

An example of a non-commutative group would be symmetry groups. The symmetry group S_3 is the group of symmetries of an equilateral triangle. It has 6 elements: 3 rotations and 3 reflections. Interestingly, S_3 is the smallest non-commutative group.

Definition 2.5: Ring

A set R with two operations $+$ and $\cdot$.

- $(R, +)$ is a commutative group
- The $\cdot$ is associative, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$, $\exists$ neutral element, $\cdot$ distributes over $+$ $a \cdot (b+c) = a \cdot b + a \cdot c$, $(b+c) \cdot a = b \cdot a + c \cdot a$

We denote by 0 the neutral element for $+$, and by 1 the neutral element for $\cdot$.

In general, $a \cdot b \neq b \cdot a$. If instead $a \cdot b = b \cdot a \forall a, b \in R$ then R is a **COMMUTATIVE RING**.

^a $+$ and $\cdot$ don't have to be the usual addition and multiplication

If R is a commutative ring, and every element different from 0 has an inverse for $\cdot$. Then we call it a **FIELD**.

Examples of rings:

$(\mathbb{Z}, +, \cdot)$ is a commutative ring. However, it is not a field as $2 \in \mathbb{Z}$ does not have an inverse for $\cdot$ in $\mathbb{Z}$.

$(\mathbb{Q}, +, \cdot)$ is a field.

Rings are generalizations for integers, and fields are generalizations for rationals.

Remember: For a group, $(a^{-1})^{-1} = a$. For a field, this means that $-(-a) = a$ and $(a^{-1})^{-1} = a$ for all $a \neq 0$. $-$ denotes the inverse for $+$, and a^{-1} the inverse for $\cdot$.

Lemma 2.6:

Let F be a field, 0 neutral element for $+$.

- $a \cdot 0 = 0 \forall a \in F$
- $a \cdot (-b) = -(a \cdot b) = (-a) \cdot b \forall a, b \in F$
- $(-a)(-b) = a \cdot b \forall a, b \in F$

Proof.

- Since 0 is neutral, $0+0=0$. By distributivity,

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

To both sides, add $-a \cdot 0$

$$0 = a \cdot 0 + a \cdot 0 - a \cdot 0 = a \cdot 0.$$

- By the second point, $-(a \cdot b) = a \cdot (-b)$. Taking the additive inverse yields $a \cdot b = -(a \cdot (-b))$. Invoking the second point yet again but with $-b$ taking the role of b , we get $-(a \cdot (-b)) = (-a) \cdot (-b)$.

□

Remark 2.7:

Can $0=1$?

If $0=1$, then

$$a = a \cdot 1 = a \cdot 0 = 0 \forall a \in F.$$

So the field would only have one element. $F = \{0\}$. From now on, we will assume that $0 \neq 1$.

Something special about the real numbers is that they are ordered.

Given X, Y sets, $X \times Y = \{(x, y) : x \in X, y \in Y\}$ is the **CARTESIAN PRODUCT**

Given P, Q sets, $P \subseteq Q = P \subset Q$ if all elements of P are also in Q . We call P a **SUBSET** of Q .

$P \not\subseteq Q$ if $P \subseteq Q$ but they are not equal. P is a **STRICT SUBSET** of Q .

$P \not\subset Q$ means P is not contained in Q .

Definition 2.8: Relation

X set, A relation is $R \subseteq X \times X$. We say that x and y are related if $(x, y) \in R$, and one writes xRy .

Symbols for relations are: $<, \leq, \sim, \cong$

Example 2.9:

Consider the set $X = \{1, 2, 3\}$. A possible relation is

$$R = \{(1, 1), (2, 2)\}.$$

A relation can be:

- **REFLEXIVE**: $x \sim x \forall x \in X$
- **TRANSITIVE**: if $x \sim y$ and $y \sim z$ implies $x \sim z$
- **SYMMETRIC**: if $x \sim y$ implies $y \sim x$
- **ANTISYMMETRIC**: if $x \sim y$ and $y \sim x$ implies $x = y$

We call an **EQUIVALENCE RELATION** a relation that is reflexive, transitive and symmetric.

If instead it is reflexive, transitive and antisymmetric, we call it a **ORDER RELATION**.

Example 2.10: Equivalence Relation

on $\mathbb{Z}$, $m \sim n$ if $m - n$ is divisible by 3. This is reflexive, transitive and symmetric.

Example 2.11: Order Relation

The classic is $x \leq y$ on $\mathbb{R}$. This is an order relation. $x < y$ however is not an order, as $x < x$ is false.

If $\sim$ is an order, we use $\leq$ as a symbol.

When combining different order relations, we have the following propositions:

Proposition 2.12:

Let R_1, R_2 be order relations on X . Then

1. $R_1 \cup R_2$ does not need to be an order relation.
2. $R_1 \cap R_2$ is an order relation.

Proof. We proof by counterexample and verification.

1. Let $X = \{1, 2\}$ and the relation $R_1 = \{(1, 1), (2, 2), (1, 2)\}$ and $R_2 = \{(1, 1), (2, 2), (2, 1)\}$. Then

$$R_1 \cup R_2 = \{(1, 1), (2, 2), (1, 2), (2, 1)\}$$

is not antisymmetric. $\nexists$

2. $R_1 \cap R_2$ is reflexive as $(x, x) \in R_1$ and $(x, x) \in R_2 \forall x \in X$.

Furthermore it is antisymmetric: if $(x, y) \in R_1 \cap R_2$ and $(y, x) \in R_1 \cap R_2$, then $(x, y) \in R_1$ and $(y, x) \in R_1$ implying $x = y$. Similarly for R_2 .

Lastly, it is transitive: if $(x, y) \in R_1 \cap R_2$ and $(y, z) \in R_1 \cap R_2$, then $(x, y) \in R_1$ and $(y, z) \in R_1$ implying $(x, z) \in R_1$. Similarly for R_2 . Thus $(x, z) \in R_1 \cap R_2$.

□

Definition 2.13:

$(F, \leq)$ is an **ORDERED FIELD** if F is a field and $\leq$ is an order relation on F such that

- Given $x, y \in F$, either $x \leq y$ or $y \leq x$
- Given $x, y, z \in F$, if $x \leq y$, then $x + z \leq y + z$
- Given $x, y \in F$, if $0 \leq x$ and $0 \leq y$, then $0 \leq x \cdot y$

Remark 2.14:

Finite fields cannot be ordered.

Proof. Exercise to the reader.

□

$(F, \leq)$ we write $y \geq x$ if $x \leq y$. Further $x < y$ if $x \leq y$ and $x \neq y$. We also say x is **NON-NEGATIVE** if $0 \leq x$ and x is **POSITIVE** if $0 < x$ and **NON-POSITIVE** if $x \leq 0$ and **NEGATIVE** if $x < 0$.

Example 2.15:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \setminus \{0\} \right\}.$$

We say

$$\frac{p}{q} \leq \frac{r}{s} \text{ if } p \cdot s \leq r \cdot q.$$

Recall that $x \in F \Rightarrow -x$ is the inverse for $+$ ($x + (-x) = 0$) and $(-x) \cdot (-y) = x \cdot y$.

Lemma 2.16:

Let $(F, \leq)$ be an ordered field. For example: (c) If $x \leq y$ and $z \leq w$, then $x + z \leq y + w$. (Same with $<$) or (f) $x^2 \geq 0$ ($x^2 = x \cdot x$) or (g) $1 > 0$.

Proof. (c) $x \leq y \Rightarrow x + z \leq y + z$ and $z \leq w \Rightarrow z + y \leq w + y$. Using transitivity yields the result.

(f) Case 1: $x \geq 0$. Then $x \cdot x \geq 0$ by definition. Case 2: $x \leq 0 \Rightarrow -x \geq 0 \Rightarrow x \cdot x = (-x) \cdot (-x) \geq 0$.

(g) $1 = 1 \cdot 1 = 1^2 \geq 0$. Also $1 \neq 0$ implying $1 > 0$.

□

Lemma 2.17:

$(F, \geq), 0, 1$

- We can define the elements $2 = 1 + 1, 3 = 2 + 1, \dots$ $-n$ is the inverse of n for $+$. Since $0 < 1 \Rightarrow 1 < 1 + 1 = 2 \Rightarrow 2 < 2 + 1 = 3 \Rightarrow \dots$. Also $0 < 1 \Rightarrow -1 < 0, -2 \leq -1, \dots$. Thus we have a sequence of elements

$$\dots < -2 < -1 < 0 < 1 < 2 < \dots \subseteq F.$$

This is exactly the integers $\mathbb{Z}$.

- Given $p, q \in F, q \neq 0$, we define $\frac{p}{q} = p \cdot q^{-1}$. Then

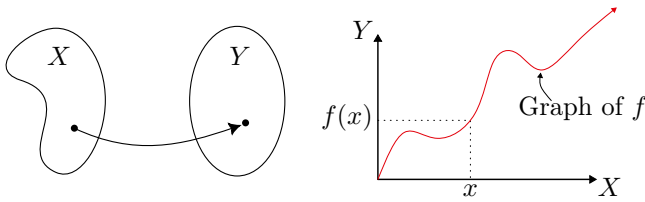
$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{Z} \setminus \{0\} \right\} \subsetneq \mathbb{Z}.$$

The inequality comes from for example $\frac{1}{2}$. Since $0 < 1 < 2 \Rightarrow \frac{1}{2} < 1$. Thus $\frac{1}{2} \notin \mathbb{Z}$.

The moral of the story is that $\mathbb{Z} \subsetneq \mathbb{Q} \subseteq F$.

Definition 2.18: Function

Given X, Y sets, a **FUNCTION** f from X to Y is an assignement of an element in Y for every element of X .



Function $f : X \rightarrow Y$

Graph of a function

We call X the **DOMAIN** of f , Y the **CODOMAIN** of f . The element $y \in Y$ to which $x \in X$ is assigned is denoted by $f(x)$.

x is called the **ARGUMENT** of f , and $y = f(x)$ is the **VALUE** of f . The set $\{(x, f(x)) : x \in X\} \subseteq X \times Y$ is called the **GRAPH** of f .

Sometimes also the notation $f : X \rightarrow Y$ and $x \mapsto f(x)$ is used.

Remark 2.19:

Two functions $f_1 : X_1 \rightarrow Y_1, f_2 : X_2 \rightarrow Y_2$ are equal if

$$\begin{aligned} X_1 &= X_2 \\ Y_1 &= Y_2 \\ f_1(x) &= f_2(x) \forall x \in X_1 \end{aligned}$$

We can define the functions sign and modulus.

Definition 2.20:

$(F, \leq)$ ordered field.

- The absolute value or the modulus is the function $|\cdot| : F \rightarrow F$ defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

- the **SIGN** is $\text{sgn} : F \rightarrow \{-1, 0, 1\}$ defined by

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}.$$

Lemma 2.21:

Let $(F, \leq)$ be an ordered field.

- $x = \text{sgn}(x) \cdot |x|, |x| = |-x|, \text{sgn}(-x) = -\text{sgn}(x)$
- $|x| \geq 0$
- $|x \cdot y| = |x| \cdot |y|$
- $|x| \geq y \Leftrightarrow -y \leq x \leq y$
- $|x + y| \leq |x| + |y|$ (Triangle inequality)
- $||x| - |y|| \leq |x - y|$ (Inverse triangle inequality)

Sometimes we write xy for $x \cdot y$ when there is no ambiguity.

Proof. [Proof of (e)] Two cases: $x \geq 0$ and $x < 0$.

- If $|x| \leq y$, since in this case $x = |x| \geq 0, -y \leq 0 \leq x \leq y$.
If $-y \leq x \leq y \Rightarrow |x| = x \leq y \Rightarrow |x| \leq y$.

□

Lec 4

When putting rationals on a number line, there will be gaps left. The classic example is $\sqrt{2}$, which cannot be expressed as a rational number. Let

$$X = \{q \in \mathbb{Q} : q < 0\} \cup \{q \in \mathbb{Q} : 0 < q^2 < 2\}.$$

$$Y = \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 > 2\}.$$

These sets are disjoint, and on the number line, X is to the left of Y , with a gap in between.

Definition 2.22:

$(F, \leq)$. We say that it is **COMPLETE** if the **COMPLETENESS AXIOM** holds.

Let $X, Y \subseteq F$ non empty, assume that $x \leq y$ for all $x \in X, y \in Y$. Then there exists $c \in F$ such that

$$x \leq c \leq y \quad \forall x \in X, y \in Y.$$

Definition 2.23: Real numbers

The **REAL NUMBERS** are any complete ordered field.

This field is the only ordered field that is complete.

Definition 2.24:

Let $f : X \rightarrow Y$ be a function.

1. f is **INJECTIVE** if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.
2. f is **SURJECTIVE** if for all $y \in Y$, there exists $x \in X$ such that $f(x) = y$.
3. f is **BIJECTIVE** if it is both injective and surjective. In this case, we denote $f^{-1} : Y \rightarrow X$ the inverse function of f , defined by $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

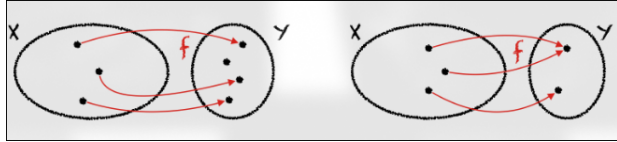


Figure 4: Injective and surjective function [1]

Definition 2.25:

$f : X \rightarrow Y$, $A \subseteq X$. Then

$$f(A) = \{y \in Y : \exists x \in A \text{ such that } y = f(x)\}$$

is the **IMAGE** of A through f .

If $B \subseteq Y$, then

$$f^{-1}(B) = \{x \in X : f(x) \in B\}$$

is the **PREIMAGE** of B through f .

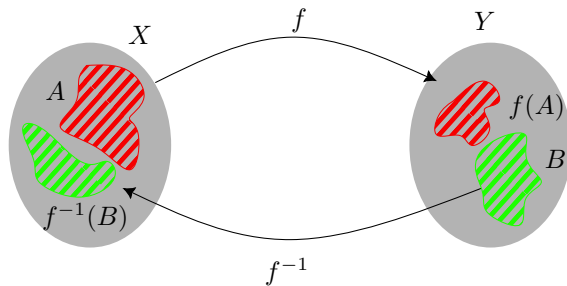


Figure 5: Image and preimage of sets

Example 2.26:

$$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto 0.$$

Then $f^{-1}(\{0\}) = \mathbb{R}$, whereas $f^{-1}(\{y\}) = \emptyset$ for $y \neq 0$.

Lemma 2.27:

X, Y finite sets with n elements. $f : X \rightarrow Y$. Then f is injective iff it is surjective.

Proof. Denote the elements of X by $x_1, \dots, x_n$.

$\Rightarrow$ Assume f is injective. Then all elements $f(x_1), \dots, f(x_n)$ are distinct. So $f(X)$ has n elements and $f(X) \subseteq Y$, Y has n elements. Therefore $f(X) = Y$ implying f is surjective.

$\Leftarrow$ We proof by contraposition. Assume f is not injective. Thus $\exists x_i \neq x_j$ such that $f(x_i) = f(x_j)$. But then $f(X) = \{f(x_1), \dots, f(x_n)\}$ has

at most $n-1$ elements. And because $f(X) \subsetneq Y$ f can not be surjective. Concluding the proof. $\square$

The lemma is not true for infinite sets.

$$f_1 : \mathbb{N} \rightarrow \mathbb{N}, n \mapsto n + 1.$$

This function is injective but not surjective ($f^{-1}(\{0\}) = \emptyset$).

$$f_2 : \mathbb{N} \rightarrow \mathbb{N}, n \mapsto \begin{cases} n = 0 & n = 0 \\ n - 1 & n \geq 1 \end{cases}.$$

f_2 is surjective but not injective ($f_2(0) = f_2(1)$).

Exercise 2.28:

Let $R_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$. Show the existence and uniqueness of the function $\sqrt{\cdot} : R_{\geq 0} \rightarrow R_{\geq 0}$ where $(\sqrt{a})^2 = a \quad \forall a \in R_{\geq 0}$.

For existence: Given $a \in R_{\geq 0}$, let $X = \{x \in R_{\geq 0} : x^2 \leq a\}$ and $Y = \{y \in R_{\geq 0} : y^2 \geq a\}$. By completeness, $\exists c \in R_{\geq 0}$ such that

$$x \leq c \leq y \quad \forall x \in X, y \in Y.$$

To proof: $c^2 = a$. (If $c^2 > a$, then $\exists \epsilon > 0$ st $(c - \epsilon)^2 > a$, so $y = c - \epsilon \in Y$ but $y < c$, contradiction. The case $c^2 < a$ is similar).

Let $a \leq b \in \mathbb{R}$. Then $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ is a **CLOSED INTERVAL**. a is called the **LEFT ENDPOINT** and b the **RIGHT ENDPOINT**. $b - a$ is the **LENGTH** of the interval.

Similarly, $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ is an **OPEN INTERVAL**.

The notations can be comined: $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ and $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$. These are **HALF-OPEN INTERVALS**.

The notation $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$ represents an **UNBOUNDED CLOSED INTERVAL**. Similarly, $(-\infty, b) = \{x \in \mathbb{R} : x \leq b\}$.

UNBOUNDED OPEN INTERVALS are defined as $(a, \infty) = \{x \in \mathbb{R} : x > a\}$ and $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$.

Intervals which include $\pm\infty$ are called **UNBOUNDED INTERVALS**. Whereas the others are **BOUNDED INTERVALS**.

Remark 2.29:

There also exists the notation $]a, b[$ for (a, b) and $]a, b]$ for $(a, b]$.

To work with intervals, we need the following definitions for sets.

Definition 2.30:

Let P, Q be sets. Then we define

$$P \cap Q = \{x : x \in P \text{ and } x \in Q\} \quad (\text{intersection}).$$

$$P \cup Q = \{x : x \in P \text{ or } x \in Q\} \quad (\text{union}).$$

$$P \setminus Q = \{x : x \in P \wedge x \notin Q\} \quad (\text{relative complement}).$$

$$P \Delta Q = (P \setminus Q) \cup (Q \setminus P) \quad (\text{symmetric difference}).$$

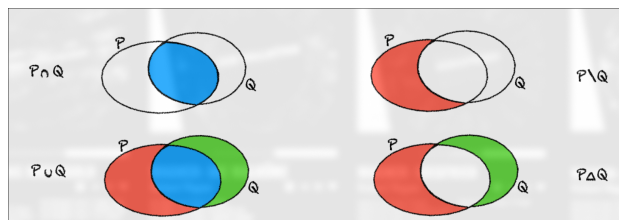


Figure 6: Set operations[1]

Given $P \subseteq X$, $P^c = X \setminus P$ is the **COMPLEMENT** of P in X . However this should be used with care, since the set X might not be obvious from the context.

Example 2.31: Complement Ambiguity

Given an interval $[a, b]$, then $[a, b]^c = (-\infty, a) \cup (b, \infty)$ might be the intuition. However, this only is true if we think about the reals. However, the notation might also be used in $\mathbb{Z}$ or $\mathbb{Q}$.

Definition 2.32: Union and Intersections of several sets

Given $\mathcal{A}$ a family of sets, we define

$$\bigcup_{A \in \mathcal{A}} A = \{x | \exists A \in \mathcal{A} : x \in A\}.$$

$$\bigcap_{A \in \mathcal{A}} A = \{x | x \in A \forall A \in \mathcal{A}\}.$$

If $\mathcal{A} = \{A_1, \dots, A_n\}$ we also write

$$\bigcup_{i=1}^n A_i \quad \text{and} \quad \bigcap_{i=1}^n A_i.$$

Example 2.33:

$\mathcal{A} = \{[x, \infty) \text{ with } x \in \mathbb{R}\}$. Then

$$\bigcup_{A \in \mathcal{A}} A = \mathbb{R}$$

$$\bigcap_{A \in \mathcal{A}} A = \emptyset$$

The second one is true since for any $y \in \mathbb{R}$, we can choose $x = y + 1$. Then $y \notin [x, \infty)$.

Lec 5

Definition 2.34:

Let $x \in \mathbb{R}$. A **NEIGHBORHOOD** of x is any set that contains an open interval I s.t. $x \in I$.

Given $\delta > 0$, the open interval $(x - \delta, x + \delta)$ is the **δ -NEIGHBORHOOD** of x .

For example, $0 \in R$.

- $[-1, 1]$ is a neighborhood of 0 as it contains $0 \in (-1, 1)$.
- $[-1, 1] \cup \mathbb{Q}$ for the same reason.

For the delta neighborhood, we have

$$(x - \delta, x + \delta) = \{y \in \mathbb{R} : |x - y| < \delta\}.$$

With the notion of neighborhood, we can define further

Definition 2.35: Open and closed sets

A set $U \subseteq \mathbb{R}$ is **OPEN** if $\forall x \in U \exists I$ open interval s.t. $x \in I \subseteq U$.

A set $V \subseteq \mathbb{R}$ is **CLOSED** if $\mathbb{R} \setminus V$ is open.

This definition is best understood with a number line. On it, one can see that open intervals are open sets, while closed intervals are closed sets. Half-open intervals are neither open nor closed, as can be seen by looking at the third example in figure 7.

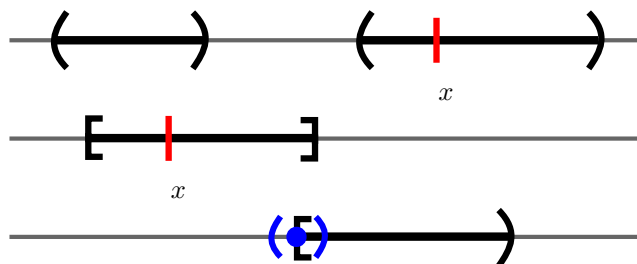


Figure 7: Open and closed sets

Theorem 2.36:

Let $U, V \subseteq \mathbb{R}$ be open. Then $U \cup V$ and $U \cap V$ are open. The same holds for arbitrary unions, but only for finite intersections.

Proof. Let U, V be open. Let $x \in U \cup V$. Wlog, assume $x \in U$. Because U is open, $\exists I$ open interval s.t. $x \in I \subseteq U \subseteq U \cup V$. Thus, $U \cup V$ is open.

Let $x \in U \cap V$. Then $x \in U$ and $x \in V$. Thus, $\exists \varepsilon_u, \varepsilon_v > 0$ s.t.

$$(x - \varepsilon_u, x + \varepsilon_u) \subseteq U, \quad (x - \varepsilon_v, x + \varepsilon_v) \subseteq V.$$

Now let $\varepsilon = \min(\varepsilon_u, \varepsilon_v)$. Then

$$(x - \varepsilon, x + \varepsilon) \subseteq (x - \varepsilon_u, x + \varepsilon_u) \cap (x - \varepsilon_v, x + \varepsilon_v) \subseteq U \cap V.$$

□

Example 2.37: Open and closed sets

$[0, 1]$ is closed in $\mathbb{R}$. Indeed, we can show that $\mathbb{R} \setminus [0, 1] = (-\infty, 0) \cup (1, \infty)$ is open. Let $x \in \mathbb{R} \setminus [0, 1]$. Wlog, assume $x \in (-\infty, 0)$. Thus, $x < 0 \Rightarrow |x| > 0$. Let $\varepsilon = \frac{|x|}{2} > 0$. By construction, is $(x - \varepsilon, x + \varepsilon) \subseteq (-\infty, 0)$. Thus, $\mathbb{R} \setminus [0, 1]$ is open, implying $[0, 1]$ is closed.

Remark 2.38:

Both $\mathbb{R}$ and $\emptyset$ are open and closed. They are the only subsets of $\mathbb{R}$ with this property.

2.2 Complex Numbers

Complex numbers can be looked at as

$$\mathbb{C} = \mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}.$$

This can be simply displayed on a plane.

We now want to define addition and multiplication on $\mathbb{C}$ to have a notion of a field. Let $z = (x, y) \in \mathbb{R}^2$, or $z = x + iy$ where x is called the **REAL PART** and y the **IMAGINARY PART**.

We first consider the map $\mathbb{R} : x \mapsto (x, 0) \in \mathbb{C}$.

Goal: Field, s.t. my operations coincide with the usual ones when considering points in $\mathbb{R}$. Also, we want $i^2 = -1$ ($(0, 1) \cdot (0, 1) = (-1, 0)$).

Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$. We let

$$z_1 \cdot z_2 = (x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1).$$

Definition 2.39:

On $\mathbb{C} = \mathbb{R}^2$, define

$$\begin{aligned} (x_1, y_1) + (x_2, y_2) &= (x_1 + x_2, y_1 + y_2) \\ (x_1, y_1) \cdot (x_2, y_2) &= (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1) \end{aligned}$$

Theorem 2.40:

$\mathbb{C}$ with $+, \cdot$ as before is a field.

Proof. We take $(0, 0)$ as neutral element for $+$, and $(1, 0)$ for $\cdot$.

The addition is immediate to check. For multiplication, we have to check

$$(x_1, y_1) \cdot ((x_2, y_2) \cdot (x_3, y_3)) = ((x_1, y_1) \cdot (x_2, y_2)) \cdot (x_3, y_3),$$

Which can be done by straightforward computation.

The neutral element for multiplication is also a simple check, similar to distributivity.

Thus $\mathbb{C}$ is a commutative ring. Given $(x, y) \in \mathbb{C} \setminus \{(0, 0)\}$, consider $\left(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2}\right)$. Then

$$\begin{aligned} (x, y) \cdot \left(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2}\right) &= \left(\frac{x^2+y^2}{x^2+y^2}, 0\right) \\ &= (1, 0). \end{aligned}$$

□

From now on, we write $x + iy$ for (x, y) , x for $x + i \cdot 0$ and iy for $0 + iy$.

Given $z, w \in \mathbb{C}$, we write zw for $z \cdot w$, if $z \neq 0$. then z^{-1} or $\frac{1}{z}$ denotes the multiplicative inverse.

For example,

$$(i)^{-1} = \frac{1}{i} = -i.$$

Definition 2.41: Complex Conjugation

Given $z = x + iy \in \mathbb{C}$, we define $\bar{z} \in \mathbb{C}$ as $x - iy$ as the **COMPLEX CONJUGATE** of z .

Lemma 2.42:

The following holds for all $z, w \in \mathbb{C}$:

1. $z \cdot \bar{z} = x^2 + y^2 \geq 0$, also $z \cdot \bar{z} = 0 \Leftrightarrow z = 0$.
2. $\overline{z + w} = \bar{z} + \bar{w}$.
3. $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$.

Proof. 1. Is a quick computation. For the second and third, let $z = x_1 + iy_1, w = x_2 + iy_2$. then

$$\begin{aligned} \overline{z + w} &= \overline{(x_1 + x_2) + i(y_1 + y_2)} \\ &= (x_1 + x_2) - i(y_1 + y_2) \\ &= (x_1 - iy_1) + (x_2 - iy_2) \\ &= \bar{z} + \bar{w}. \end{aligned}$$

Similarly, for the third. □

As an exercise, Given $z = x + iy$, write $x = \Re(z)$ and $y = \Im(z)$. Then

$$\Re(z) = \frac{z + \bar{z}}{2}, \quad \Im(z) = \frac{z - \bar{z}}{2i}.$$

Recall that on $\mathbb{R}$, given $x \in \mathbb{R}$, $x \mapsto |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$.

Because $\mathbb{C}$ is not an ordered field we need something different. The modulus represents the distance to 0. We want to implement something similar on $\mathbb{C}$. By the pythagorean theorem, the distance to 0 is $\sqrt{x^2 + y^2}$.

Definition 2.43:

Given $z \in \mathbb{C}$

$$|z| = \sqrt{z \cdot \bar{z}} = \sqrt{x^2 + y^2}.$$

Where $\sqrt{\cdot}$ is the square root defined before.

If $z = x \in \mathbb{R}$, then the two definitions of $|\cdot|$ coincide.

Some properties of the modulus:

$$|zw| = \sqrt{zw \cdot \overline{zw}} = \sqrt{zw \cdot \bar{z} \cdot \bar{w}} = \sqrt{z \cdot \bar{z}} \cdot \sqrt{w \cdot \bar{w}} = |z| \cdot |w|.$$

$$z^{-1} = \frac{\bar{z}}{|z|^2} \quad \text{for } z \neq 0.$$

Theorem 2.44: Triangle Inequality

For all $z, w \in \mathbb{C}$,

$$|z + w| \leq |z| + |w|.$$

Proof. We have

$$\begin{aligned} |z + w|^2 &= (x_1 + x_2)^2 + (y_1 + y_2)^2 \\ &= x_1^2 + y_1^2 + x_2^2 + y_2^2 + 2(x_1x_2 + y_1y_2) \\ &= |z|^2 + |w|^2 + 2(x_1x_2 + y_1y_2) \\ &\leq |z|^2 + |w|^2 + 2|z| \cdot |w| \\ &= (|z| + |w|)^2. \end{aligned}$$

However we still have to show $x_1x_2 + y_1y_2 \leq |z| \cdot |w|$. □

Lec 6

Lemma 2.45: Cauchy-Schwarz inequality

$z = x_1 + iy_1, w = x_2 + iy_2 \in \mathbb{C}$. Then

$$x_1x_2 + y_1y_2 \leq |z||w|.$$

Proof. Look at $|z|^2|w|^2 - (x_1x_2 + y_1y_2)^2$. This is equal to

$$\begin{aligned} &= (x_1^2 + y_1^2)(x_2^2 + y_2^2) - (x_1x_2 + y_1y_2)^2 \\ &= x_1^2y_2^2 + x_2^2y_1^2 - 2x_1x_2y_1y_2 \\ &= (x_1y_2 - x_2y_1)^2 \geq 0 \Rightarrow |z|^2|w|^2 \geq (x_1x_2 + y_1y_2)^2 \end{aligned}$$

Taking square roots gives

$$|z||w| \geq |x_1x_2 + y_1y_2| \geq x_1x_2 + y_1y_2.$$

□

In general, $|z - w|$ represents the length of the segment between z and w .

Definition 2.46: Open and closed disks

Given $z \in \mathbb{C}$ and $r > 0$, we define

$$B(z, r) = \{w \in \mathbb{C} \mid |w - z| < r\},$$

as the **OPEN DISK** of radius r centered at z .

Furthermore, we define

$$\overline{B(z, r)} = \{w \in \mathbb{C} \mid |w - z| \leq r\},$$

as the **CLOSED DISK** of radius r centered at z .

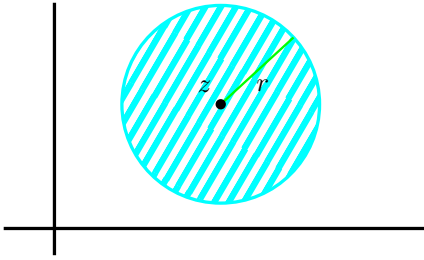


Figure 8: Disks in $\mathbb{C}$.

Notice that this definition is a generalisation of open and closed intervals, as it reduces to them by looking at $B(z, r) \cap \mathbb{R} = (z - r, z + r)$.

This definition can be used to define open and closed sets in $\mathbb{C}$.

Definition 2.47:

$U \subseteq \mathbb{C}$ is **OPEN** if for all $z \in U \exists r > 0$, such that $B(z, r) \subseteq U$.

A set $C \subseteq \mathbb{C}$ is **CLOSED** if $\mathbb{C} \setminus C$ is open.

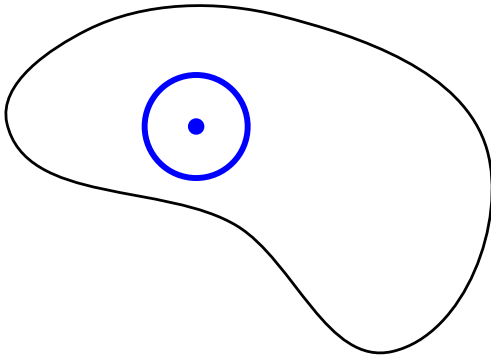


Figure 9: Open and closed sets in $\mathbb{C}$.

Again sets do not have to be either open or closed. For example,

$$A = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 < y < 1\}.$$

2.3 Maximum and Supremum

Let us return to the real numbers for the moment.

Definition 2.48:

$X \subseteq \mathbb{R}$. We say that

- X is **BOUNDED ABOVE** if $\exists s \in \mathbb{R}$ such that $x \leq s \forall x \in X$. Such an s is called an **UPPER BOUND** of X .

If such an s belongs to X , then s is called the **MAXIMUM** of X , denoted $s = \max(X)$.

- Analogous definitions hold for **BOUNDED BELOW**, **LOWER BOUND** and **MINIMUM**.
- A set is **BOUNDED** if it is both bounded above and below.

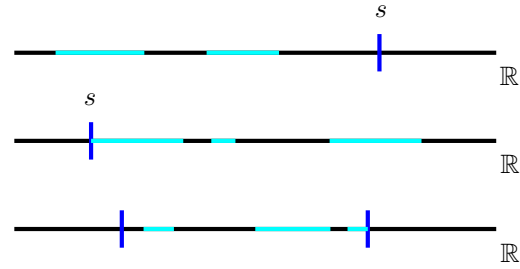


Figure 10: Bounded sets.

Remark 2.49:

The maximum, if it exists, is unique.

Proof. Indeed, if $s_1, s_2 \in X$ are maxima of X , then

$$s_1 \leq s_2 \text{ and } s_2 \leq s_1 \Rightarrow s_1 = s_2.$$

□

Definition 2.50: Supremum

Let $X \subseteq \mathbb{R}$. Define

$$A = \{a \in \mathbb{R} \mid a \geq x \forall x \in X\}.$$

This is the set containing all upper bounds of X .

If A has a minimum, then we call it the **SUPRENUM** of X , and denote it by $\sup(X)$.

$s = \sup(X)$ means that $x \leq s \forall x \in X$ and that $\forall t < s \exists x \in X$ such that $x > t$.

Remark 2.51:

If X has a maximum, then $\max(X) = \sup(X)$.

Example 2.52:

A few examples:

- $[a, b]$ is bounded with $\min([a, b]) = a$ and $\max([a, b]) = b$.
- (a, b) is bounded, but has no minimum or maximum. However, $\sup((a, b)) = b$.

Probably one of the most important properties of $\mathbb{R}$ is the following:

Theorem 2.53: Existence of Supremum

Let $X \subseteq \mathbb{R}$ be bounded above. Then $\sup(X)$ exists.

Proof. Let $A = \{a \in \mathbb{R} \mid a \geq x \forall x \in X\}$ be the set of upper bounds. Since X is bounded above, A is nonempty. By completeness, $\exists c \in \mathbb{R}$ such that

$$x \leq c \leq a \forall x \in X, a \in A.$$

The first inequality means that c is an upper bound of X , instead looking at the second inequality, we see that c is the least upper bound. So $c = \min(A) = \sup(X)$. $\square$

If X is bounded below, the largest lower bound is called **INFIMUM**, written $\inf(X)$.

Proposition 2.54:

Given $X, Y \subseteq \mathbb{R}$, we can define the sets

$$X + Y = \{x + y \mid x \in X, y \in Y\}$$

$$XY = \{xy \mid x \in X, y \in Y\}$$

$$1. \quad \sup(X \cup Y) = \max(\sup(X), \sup(Y)).$$

$$2. \text{ If } X \cap Y \neq \emptyset, \text{ then } \sup(X \cap Y) \leq \min(\sup(X), \sup(Y)).$$

$$3. \quad \sup(X + Y) = \sup(X) + \sup(Y).$$

$$4. \text{ If } X, Y \subseteq \mathbb{R}_{\geq 0}, \text{ then } \sup(XY) = \sup(X) \cdot \sup(Y).$$

Proof. [Proof of 3.] Let $x_0 = \sup(X)$ and $y_0 = \sup(Y)$.

Given $z \in X + Y$, by definition $\exists x \in X \wedge y \in Y$ such that $z = x + y$. Therefore, $z = x + y \leq x_0 + y_0$. This implies that $x_0 + y_0$ is an upper bound for $X + Y$, which means that $\sup(X + Y) \leq x_0 + y_0$.

To assume equality, assume by contradiction that $x_0 + y_0 - \sup(X + Y) = \epsilon > 0$.

Since $x_0 = \sup(X)$, $\exists \bar{x} \in X : \bar{x} > x_0 - \frac{\epsilon}{2}$. Similarly, $\exists \bar{y} \in Y : \bar{y} > y_0 - \frac{\epsilon}{2}$.

Therefore, $\sup(X + Y) \geq \bar{x} + \bar{y} > x_0 + y_0 - \epsilon = \sup(X + Y)$, which is a contradiction. $\frac{1}{2}$

$\square$

We denote $\overline{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$ for the **EXTENDED REAL LINE**. The rule is, that for all $x \in \mathbb{R}$, $-\infty < x < \infty$. Furthermore,

- $\infty + x = \infty$
- $-\infty + x = -\infty$
- If $x > 0$, $x \cdot \infty = \infty$, $x \cdot -\infty = -\infty$

But, $\infty - \infty$ and $0 \cdot \infty$ NO, VERBOTEN!!!

Definition 2.55:

$X \subseteq \mathbb{R}$ and X not bounded above $\Rightarrow \sup(X) = \infty$.

Lets consider $X = \emptyset$. Then $\sup(\emptyset) = -\infty$.

Theorem 2.56: Archimedian Principle

Given $x \in \mathbb{R} \exists n \in \mathbb{Z}$ such that $n \leq x < n + 1$

Proof. Consider first $x \geq 0$. Let $E = \{n \in \mathbb{Z} \mid n \leq x\}$. Since $x \geq 0$, $0 \in E$ so $E \neq \emptyset$. Also E is bounded above. So $\exists s_0 = \sup E \in \mathbb{R}$.

Since s_0 is the least upper bound, so $s_0 \leq x$ because x is an upper bound and $\exists n_0 \in E$ such that $s_0 - 1 < n_0$, otherwise $s_0 - 1$ would be an upper bound.

The latter implies $s_0 < n_0 + 1$. Since $m \leq s_0 \forall m \in E$, $m < n_0 + 1 \forall m \in E \subseteq \mathbb{Z}$. As $m, n_0 \in \mathbb{Z}$, this implies $m \leq n_0$.

Since $n_0 \in E$, this implies that $n_0 = \max(E)$, implying $n_0 = s_0$.

Also, $n_0 = \max(E) \Rightarrow n_0 + 1 \notin E \Rightarrow n_0 + 1 > x$.

So recalling that $s_0 \leq x$, we have

$$n_0 = s_0 \leq x < n_0 + 1.$$

If $x < 0$, apply the result to $-x$. Then $\exists m \in \mathbb{Z}$, such that

$$m \leq -x < m + 1 \Rightarrow -m - 1 < x \leq -m.$$

If $x = -m$, take $n = -m$:

$$n = -m = x < n + 1.$$

If $x < -m$, take $n = -m - 1$:

$$n < x < n + 1.$$

For uniqueness, fix an x , assume $\exists n_1, n_2 \in \mathbb{Z}$ such that

$$n_1 \leq x < n_1 + 1$$

$$n_2 \leq x < n_2 + 1.$$

Combining the inequalities diagonally, we get

$$n_1 < n_2 + 1 \Rightarrow n_1 \leq n_2$$

$$n_2 < n_1 + 1 \Rightarrow n_2 \leq n_1.$$

This however implies $n_1 = n_2$. $\square$

Definition 2.57:

Given $x \in \mathbb{R}$, let n be given as before ($n \in \mathbb{Z}, n \leq x < n + 1$). Define

$$n = \lfloor x \rfloor,$$

as the **INTEGER PART** of x .

Furthermore, let

$$\{x\} = x - \lfloor x \rfloor \in [0, 1),$$

be the **FRACTIONAL PART** of x .

Corollary 2.58:

For every $\epsilon > 0, \exists n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$.

Proof. If $\epsilon \geq 1$, take $n = 1$.

If $\epsilon < 1$, then consider $\frac{1}{\epsilon} \geq 1$. By Archimedian principle, $\exists m \in \mathbb{Z}$ such that $m \leq \frac{1}{\epsilon} < m + 1$. Let $n = m + 1 \in \mathbb{N}$. Then

$$\frac{1}{\epsilon} < n \Leftrightarrow \frac{1}{n} < \epsilon.$$

$\square$

Corollary 2.59:

$\forall a < b \in \mathbb{R}, \exists r \in \mathbb{Q}$ such that $a < r < b$.

Proof. Let $\epsilon = b - a$. By the corollary 2.58, $\exists m \geq 1$ such that $\frac{1}{m} < \epsilon$. Consider $x = ma$. By the Archimedian principle, $\exists n \in \mathbb{Z}$ such that $n \leq ma < n + 1$. Then

$$\frac{n}{m} \leq a < \frac{n+1}{m} \leq a + \frac{1}{m} < a + \epsilon = b.$$

□ If $a_0 < 0$, consider $-a_0$, find c and consider $-c$.

Definition 2.60:

A set $X \subseteq \mathbb{R}$ is **DENSE** in every open interval contains an element of X .

Proposition 2.61:

$\mathbb{Q}$ is dense in $\mathbb{R}$.

Proof. Let $\varepsilon > 0$. Choose $x \in \mathbb{R}$ and let $q \in \mathbb{N}_{\geq 0}$ such that $\frac{1}{q} < \varepsilon$. By Archimedian principle, $\exists p \in \mathbb{Z}$ such that $p \leq qx < p+1$. Dividing by q , we have

$$\frac{p}{q} \leq x < \frac{p+1}{q} < x + \frac{1}{q} < x + \varepsilon.$$

Thus $|x - \frac{p}{q}| < \varepsilon$.

□

Exercise 2.62:

For every $\varepsilon > 0 \exists n \geq 1$ such that

$$\frac{1}{10^n} < \varepsilon.$$

(This uses that $10^n < 10^{n+1}$)

2.3.1 Decimal Representation of Real Numbers

We have a sequence of numbers $a_0, a_1, a_2, \dots$ where

$$a_0 \in \mathbb{Z}, a_i \in \{0, 1, 2, \dots, 9\} \forall i \geq 1.$$

Start from $a_0, a_1, a_2, \dots$ and assume $a_0 \geq 0$.

We define $x_0 = a_0$ and $y_0 = a_0 + 1$. Furthermore

$$\begin{aligned} x_1 &= a_0 + \frac{a_1}{10} & y_1 &= a_0 + \frac{a_1 + 1}{10} \\ x_2 &= a_0 + \frac{a_1}{10} + \frac{a_2}{10^2} & y_2 &= x_2 + \frac{1}{10^2} \\ x_n &= \sum_{\kappa=0}^n a_{\kappa} 10^{-\kappa} & y_n &= x_n + 10^{-n}. \end{aligned}$$

From this, we have

$$x_0 < x_1 < x_2 < \dots < x_n < y_n < \dots < y_2 < y_1 < y_0.$$

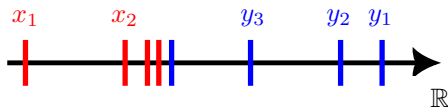


Figure 11: Decimal construction of real numbers.

Define $X = \{x_0, x_1, \dots\}$ and $Y = \{y_0, y_1, \dots\}$. Then we have $x \leq y \forall x \in X, y \in Y$. So by completeness axiom, $\exists z \in \mathbb{R}$ such that in particular $x_n \leq z \leq y_n \forall n \geq 0$.

Notice that c is unique. As if c and d both satisfy this, then

$$x_n \leq c, d \leq y_n \Rightarrow |c - d| \leq y_n - x_n = 10^{-n} \Rightarrow c = d.$$

Viceversa, given $x \in \mathbb{R}, x \geq 0$, let

$$a_0 = \lfloor x \rfloor, a_n = \lfloor 10^n x \rfloor - 10 \lfloor 10^{n-1} x \rfloor, n \geq 1.$$

Remark 2.63:

This is not a 1:1 correspondence, as

$$0.9999999 \dots = 1.000000 \dots$$

This only happens with infinite tails of 9s. So for example, instead of writing $0.5399 \dots$, we write 0.54 .

Definition 2.64: Power Set

Given X , define $\mathcal{P}(X) = \{Y \mid Y \subseteq X\}$ as the **POWER SET** of X .

It is the set of all subsets of X .

Definition 2.65:

X, Y sets. We write

- $X \sim Y$ (X and Y have the same cardinality) if $\exists f : X \rightarrow Y$ bijective,
- If $\exists f : X \rightarrow Y$ injective, we write $|X| \lesssim |Y|$,
- X is **COUNTABLE** if $\exists f : X \rightarrow \mathbb{N}$ bijective,
- X is **UNCOUNTABLE** if X infinite and not countable.

Theorem 2.66:

If $X \lesssim Y$ and $Y \lesssim X$, then $X \sim Y$.

Example 2.67: Natural and Even Numbers

Given the sets

$$\mathbb{N} = \{0, 1, 2, 3, \dots\} \text{ and } \mathbb{N}_{\text{even}} = \{0, 2, 4, 6, \dots\},$$

we have $\mathbb{N} \sim \mathbb{N}_{\text{even}}$ as the function

$$\begin{aligned} f : \mathbb{N} &\longrightarrow \mathbb{N}_{\text{even}} \\ n &\longmapsto f(n) = 2n. \end{aligned}$$

Example 2.68: Integers and Naturals

Given the sets $\mathbb{N}, \mathbb{Z}$, we clearly have $\mathbb{N} \lesssim \mathbb{Z}$ by the function $f : \mathbb{N} \rightarrow \mathbb{Z}, f(n) = n$. For the other direction, consider

$$\begin{aligned} f : \mathbb{Z} &\longrightarrow \mathbb{N} \\ n &\longmapsto f(n) = \begin{cases} 2n & n \geq 0 \\ -2n - 1 & n < 0 \end{cases}. \end{aligned}$$

Thus there must exist a bijection $\mathbb{Z} \sim \mathbb{N}$.

For the set of fractions, we have the following properties:

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{N} \setminus \{0\} \right\}.$$

The map

$$f : \mathbb{Q} \longrightarrow \mathbb{Z} \times \mathbb{N}$$

$$\frac{p}{q} \longmapsto f\left(\frac{p}{q}\right) = (p, q).$$

is injective, thus $\mathbb{Q} \lesssim \mathbb{Z} \times \mathbb{N} \setminus \{0\}$. Thus also $\mathbb{Q} \lesssim \mathbb{N} \times \mathbb{N}$.

We want to show that $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$. For this consider the following map:

$$f : \mathbb{N} \times \mathbb{N} \longrightarrow \mathbb{N}$$

$$\begin{array}{ll} (0, 0) \mapsto 0 & (1, 1) \mapsto 4 \\ (1, 0) \mapsto 1 & (2, 0) \mapsto 5 \\ (0, 1) \mapsto 2 & (3, 0) \mapsto 6 \\ (0, 2) \mapsto 3 & (2, 1) \mapsto 7 \end{array}$$

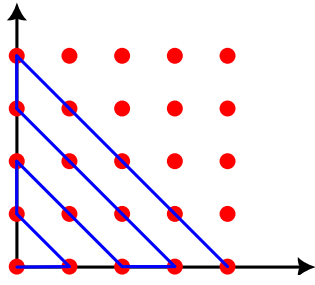


Figure 12: Mapping $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$

This map shows, that $\mathbb{N} \times \mathbb{N} \lesssim \mathbb{N}$. But clearly also $\mathbb{N} \lesssim \mathbb{N} \times \mathbb{N}$. Thus $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$, implying

$$\mathbb{Q} \lesssim \mathbb{N}.$$

When considering the set of real numbers $\mathbb{R}$, there exist more real numbers than natural numbers, i.e. $\mathbb{R} \not\sim \mathbb{N}$.

Argument 1: We proof $\nexists f : \mathbb{N} \rightarrow [0, 1)$ surjective. Assume there exists such a function. Then we can write all numbers in $[0, 1]$ as

$$\begin{aligned} r_0 &= 0, a_{00}a_{01}a_{02}a_{03} \dots \\ r_1 &= 0, a_{10}a_{11}a_{12}a_{13} \dots \\ r_2 &= 0, a_{20}a_{21}a_{22}a_{23} \dots \\ &\vdots \end{aligned}$$

Now consider the number

$$r = 0, b_0b_1b_2b_3 \dots \quad b_i = \begin{cases} 1 & a_{ii} \neq 1 \\ 2 & a_{ii} = 1 \end{cases}.$$

As this number differs from every r_i at the i -th decimal place, it follows that $r \neq r_i$ for all $i \in \mathbb{N}$. Thus $r \notin \{r_i \mid i \in \mathbb{N}\}$, contradicting the assumption that f is surjective.

2.4 Sequences of Real Numbers

Informally, a sequence is a list of numbers, indexed by $\mathbb{N}$. A more precise definition is as follows

Definition 2.69: Sequence

A **SEQUENCE** is a function

$$a : \mathbb{N} \longrightarrow \mathbb{R}, n \mapsto a_n.$$

Here a_n is the n -th element of the sequence. We often write $(a_n)_{n=0}^\infty$ or $(a_n)_{n \in \mathbb{N}}$ or $(a_n)_{n \geq 0}$ for the sequence.

In practice, we often use $x_0, x_1, x_2, \dots$ or x_n instead of a_n .

Definition 2.70: Constant Sequences

A sequence is **CONSTANT** if $x_i = x_j \forall i, j$.

A sequence $(x_n)_{n \geq 0}$ is **EVENTUALLY CONSTANT** if

$$\exists N \in \mathbb{N} : x_i = x_j \forall i, j \geq N.$$

Example 2.71: Sequences

$(x_n)_{n \geq 0}$ is $x_0 = 0, x_n = \frac{1}{n}$ for $n \geq 1$.

Another example is $(x_n)_{n \geq 0}, x_n = (-1)^n$.

Tip 2.72:

The latter sequence, $x_n = (-1)^n$, can often be used as a counterexample.

Definition 2.73: Limit of a Sequence

Let $(x_n)_{n \geq 0}$ be a sequence. We say that it **CONVERGES** if

$$\exists A \in \mathbb{R} : \forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n \geq N : |x_n - A| < \varepsilon.$$

In this case, $A = \lim_{n \rightarrow \infty} x_n$ is called the **LIMIT** of the sequence.

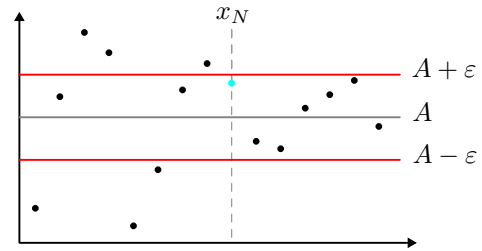


Figure 13: Definition of a Limit

Example 2.74:

$$x_n = \begin{cases} 0 & \text{if } n = 0 \\ \frac{1}{n} & \text{if } n \geq 1 \end{cases}.$$

We claim that $A = 0$ is the limit.

We fix $\varepsilon > 0$. By archimedian principle, $\exists N : 1/N < \varepsilon$. Then $\forall n \geq N$, we have

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Proposition 2.75:

The limit, if it exists, is unique.

Proof. Assume $(x_n)_{n \geq 0}$ has two limits $A, B \in \mathbb{R}$, with $A \neq B$. Let $\varepsilon = \frac{|A-B|}{3}$. The 3 is here so the delta neighborhoods around A and B do not overlap.

By the definition of the limit, $\exists N_A \in \mathbb{N}$, such that $|x_n - A| < \varepsilon$ for all $n \geq N_A$. Similarly, $\exists N_B \in \mathbb{N}$ such that $|x_n - B| < \varepsilon$ for all $n \geq N_B$.

Take $N = \max(N_A, N_B)$. Then,

$$\begin{aligned} 3\varepsilon &= |A - B| \\ &\leq |A - x_n| + |x_n - B| \\ &< \varepsilon + \varepsilon = 2\varepsilon. \end{aligned}$$

This is a contradiction, thus the limit is unique. $\square$

If we return to the example $x_n = (-1)^n$, we see that this sequence has no limit, as it does not approach any number.

We can define a subsequence which informally speaking just takes some subset of the sequence.

$$x_0, x_1, x_2, \dots \rightarrow x_0, x_1, x_4, x_{20}, x_{79}, \dots$$

Definition 2.76: Subsequence

Given $(x_n)_{n \geq 0}$, a **SUBSEQUENCE** is a sequence of the form

$$(x_{n_k})_{k \geq 0},$$

where n_k is a strictly increasing sequence of natural numbers, that is $n_{k+1} > n_k$ for all k .

Remark 2.77:

Since $n_0 \geq 0$ and $n_{k+1} > n_k$, one can prove by induction that $n_k \geq k$.

Exercise 2.78:

Let $(x_n)_{n \geq 0}$ be a sequence such that $\lim_{n \rightarrow \infty} x_n = A$. Then for every subsequence $(x_{n_k})_{k=0}^\infty$, show that

$$\lim_{k \rightarrow \infty} x_{n_k} = A.$$

Remark 2.79:

A sequence does not need to start at $n = 1$. For example, we can denote

$$\{x_1, x_2, x_3, \dots\} \leftrightarrow \{x_n\}_{n=1}^\infty.$$

This might be useful for sequences as $x_n = \frac{1}{n}, n \geq 1$.

However, there need to be infinitely many elements in the sequence.

Definition 2.80: Accumulation point

$(x_n)_{n=0}^\infty$. We say $A \in \mathbb{R}$ is an **ACCUMULATION POINT** if

$$\forall \varepsilon > 0, \forall N \in \mathbb{N} \exists n \geq N : |x_n - A| < \varepsilon.$$

Remark 2.81:

A limit is an accumulation point.

Proposition 2.82: Bolzano-Weierstrass

$A \in \mathbb{R}$ is an accumulation point $\Leftrightarrow \exists$ subsequence $(x_{n_k})_{k=0}^\infty$ such that $A = \lim_{k \rightarrow \infty} x_{n_k}$.

Proof. $\Rightarrow$: The idea of the proof is to cherry-pick the elements of the sequence that are close to A .

Assume $A \in \mathbb{R}$ is an accumulation point. First, apply the definition of accumulation point with $\varepsilon = 1$ and $N = 1$:

$$\exists n_0 \geq 1 : |x_{n_0} - A| < 1.$$

Second, apply the definition of accumulation point with $\varepsilon = \frac{1}{2}$ and $N = n_0 + 1$:

$$\exists n_1 \geq n_0 + 1 : |x_{n_1} - A| < \frac{1}{2}.$$

In general, given n_{k-1} , apply the definition of accumulation point with $\varepsilon = 2^{-k}$ and $N = n_{k-1} + 1$:

$$\exists n_k \geq n_{k-1} + 1 : |x_{n_k} - A| < 2^{-k}.$$

Now, given $\varepsilon > 0$, take $N \in \mathbb{N}$ such that $2^{-N} < \varepsilon$. Then

$$|x_{n_k} - A| < 2^{-k} \leq 2^{-N} < \varepsilon, \quad \forall k \geq N.$$

Thus $\lim_{k \rightarrow \infty} x_{n_k} = A$.

$\Leftarrow$: Fix $\varepsilon > 0$ and $N \in \mathbb{N}$. Since $A = \lim_{k \rightarrow \infty} x_{n_k} \exists N_0$ such that

$$|x_{n_k} - A| < \varepsilon, \quad \forall k \geq N_0.$$

Choose $\kappa = \max\{N_0, N\}$. Then $|x_{n_\kappa} - A| < \varepsilon$ and $n_\kappa \geq \kappa \geq N$. $\square$

Corollary 2.83:

$A \in \mathbb{R}$ accumulation point of $(x_n)_{n=0}^\infty$. Then $\forall \varepsilon > 0$ infinitely many $n \in \mathbb{N}$ such that

$$x_n \in (A - \varepsilon, A + \varepsilon).$$

Proof. A accumulation point $\Rightarrow \exists (x_{n_k})_{k=0}^\infty$ such that $A = \lim_{k \rightarrow \infty} x_{n_k}$. Fix $\varepsilon > 0$. Then $\exists N \in \mathbb{N}$ such that $|x_{n_k} - A| < \varepsilon$ for all $k \geq N$. But this means that there are infinitely many elements of the subsequence $(x_{n_k})_{k=0}^\infty$ in the interval $(A - \varepsilon, A + \varepsilon)$. $\square$

Example 2.84:

Consider the sequence $x_n = (-1)^n$. Then -1 and 1 are accumulation points.

Proposition 2.85: Limit computations

Consider two sequences $(x_n)_{n=0}^\infty$ and $(y_n)_{n=0}^\infty$ with limits A, B respectively.

- (1) The sequence $(x_n + y_n)_{n=0}^\infty$ converges to $A + B$.
- (2) The sequence $(x_n \cdot y_n)_{n=0}^\infty$ converges to AB .
- (3) Given $\alpha \in \mathbb{R}$, the sequence $(\alpha x_n)_{n=0}^\infty$ converges to αA .
- (4) If $x_n \neq 0 \forall n$ and $A \neq 0$, then the sequence $(x_n^{-1})_{n=0}^\infty$ converges to A^{-1} .

Proof. (1) Fix $\varepsilon > 0$. Since $\lim x_n = A, \lim y_n = B, \exists N_A, N_B \in \mathbb{N}$ such that

$$|x_n - A| < \frac{\varepsilon}{2}, \quad \forall n \geq N_A$$

and

$$|y_n - B| < \frac{\varepsilon}{2}, \quad \forall n \geq N_B.$$

Therefore, if $N = \max\{N_A, N_B\}$, then

$$|(x_n + y_n) - (A + B)| \leq |x_n - A| + |y_n - B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \forall n \in \mathbb{N}.$$

In the last step we used the triangle inequality.

(4) Since $A \neq 0$, we can apply the definition of the limit with $\varepsilon = \frac{|A|}{2}$. Then $\exists N_0 \in \mathbb{N}$ such that

$$|x_n - A| < \frac{|A|}{2}, \quad \forall n \geq N_0.$$

Therefore, $|x_n| = |x_n - A + A| \geq |A| - |x_n - A|$ by triangle inequality. Then

$$|A| - |x_n - A| > |A| - \frac{|A|}{2} = \frac{|A|}{2} \forall n \geq N_0.$$

This means that all x_n are bounded below by $\frac{|A|}{2}$ for $n \geq N_0$. We do this so that we don't get too close to 0.

Therefore, for $n \geq N_0$,

$$|x_n^{-1} - A^{-1}| = \frac{|x_n - A|}{|x_n||A|} \leq \frac{2}{|A|^2} |x_n - A|.$$

Where we used that $|x_n| \geq \frac{|A|}{2} \Leftrightarrow \frac{1}{|x_n|} \leq \frac{2}{|A|}$.

Now, fix $\varepsilon > 0$. Choose $N_1 \in \mathbb{N}$ such that

$$|x_n - A| < \frac{|A|^2}{2} \varepsilon, \quad \forall n \geq N_1.$$

Then for $N = \max\{N_0, N_1\}$

$$|x_n^{-1} - A^{-1}| \leq \frac{2}{|A|^2} |x_n - A| < \varepsilon, \quad \forall n \geq N.$$

□

Remark 2.86:

The other direction does NOT hold in general. For example, consider $a_n = (-1)^n, b_n = (-1)^{n+1}$. Then a_n, b_n do not converge, but $a_n + b_n = 0$ converges to 0.

Proposition 2.87: Limits with inequalities

$(x_n), (y_n)$ with limits A, B .

- (1) If $A < B$, then $\exists N$ such that $x_n < y_n \forall n \geq N$.
- (2) If $\exists N$ such that $x_n \leq y_n \forall n \geq N$, then $A \leq B$.

The strictness of (1) is important!

The proof to (1) is essentially the same as the proof to the uniqueness of limits.

Proof. Let $\varepsilon = \frac{B-A}{3}$ in order that $A + \varepsilon < B - \varepsilon$. For $n \geq N_A$, we have $|x_n - A| < \varepsilon$ and for $n \geq N_B$, we have $|y_n - B| < \varepsilon$. Therefore, for $N = \max\{N_A, N_B\}$,

$$x_n < A + \varepsilon < B - \varepsilon < y_n, \quad \forall n \geq N.$$

For (2), assume by contradiction that $A > B$. Then we can apply (1) with x_n and y_n reversed to get $\exists N_0 \in \mathbb{N}$ such that $x_n > y_n \forall n \geq N_0$. However, this contradicts $x_n \leq y_n \forall n \geq N$. □

Example 2.88:

Consider $x_n = -\frac{1}{n}$ and $y_n = \frac{1}{n}$. Then $\lim x_n = 0 = \lim y_n$. From this, we see that $x_n < y_n \not\Rightarrow A < B$.

Lemma 2.89: Sandwich lemma

$(x_n), (y_n), (z_n), \exists N$ such that

$$x_n \leq y_n \leq z_n, \quad \forall n \geq N.$$

Suppose x_n, z_n converge to the same limit $A \in \mathbb{R}$. Then

$$\lim_{n \rightarrow \infty} y_n = A.$$

Exercise 2.90:

Compute the following limits:

1. $a_n = (1 - \frac{1}{n^2})^n$.
2. $b_n = \frac{n^n}{2^{n^2}}$.
3. $c_n = \sqrt[n]{n^2 - n + 1}$.
4. $d_n = \sum_{k=1}^n \frac{k}{n^2 + k}$.

Lec 10

Definition 2.91: Bounded Sequences

A sequence $(x_n)_{n \geq 0}$ is **BOUNDED** if $\exists M > 0$ such that $|x_n| \leq M \forall n \in \mathbb{N}$.

Proposition 2.92:

If (x_n) is convergent, then it is bounded.

Proof. Let $\lim_{n \rightarrow \infty} x_n = A$. Fix $\varepsilon = 1$. Then $\exists N$ such that

$$|x_n - A| < 1, \quad \forall n \geq N.$$

Therefore, by triangle inequality,

$$|x_n| = |x_n - A + A| \leq |x_n - A| + |A| < |A| + 1.$$

We define $M := \max\{|x_0|, |x_1|, \dots, |x_{N-1}|, |A| + 1\}$. Then $|x_n| \leq M \forall n \in \mathbb{N}$. □

In general, sequences do not need to be ordered, they can be anywhere. As an order is quite useful, this motivates the following definition.

Definition 2.93: Monotone sequences

(x_n) is

- **INCREASING** if $m > n \Rightarrow x_m \geq x_n$,
- **STRICTLY INCREASING** if $m > n \Rightarrow x_m > x_n$,
- **DECREASING** if $m > n \Rightarrow x_m \leq x_n$,
- **STRICTLY DECREASING** if $m > n \Rightarrow x_m < x_n$.

Theorem 2.94:

A monotone sequence $(x_n)_{n \geq 0}$ converges iff it is bounded.

More precisely, let $X = \{x_0, x_1, \dots, x_n\} \subseteq \mathbb{R}$. If (x_n) is increasing, then $\lim_{n \rightarrow \infty} x_n = \sup X$.

If (x_n) is instead decreasing, then $\lim_{n \rightarrow \infty} x_n = \inf X$.

Proof. $\Rightarrow$: Convergent sequences are always bounded. (Prop. 2.92)

$\Leftarrow$: Let (x_n) be increasing and bounded. (x_n) bounded implies that X is bounded, which implies that $A = \sup X$ exists.

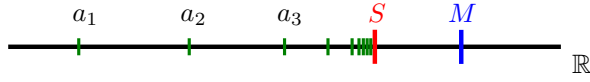


Figure 14: Bounded monotone sequences

By definition of supremum:

- (i) $x \leq A \forall n \in \mathbb{N}$ because A is an upper bound of X .
- (ii) $\forall \varepsilon > 0, \exists x_N \in X$ such that $x_n > A - \varepsilon$ (Since A is the smallest upper bound).

Then, given $\varepsilon > 0$, by (i) and (ii), we have for $n \geq N$

$$A - \varepsilon < x_n \leq x_N \leq A < A + \varepsilon.$$

Which implies that A is the limit of (x_n) . $\square$

Example 2.95: Herons Algorithm

Show that the following sequence can be used to approximate $\sqrt{2}$:

$$x_0 = 2, \quad x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right).$$

Solution.

For the limit x , it holds that

$$x = \frac{1}{2} \left(x + \frac{2}{x} \right) \Rightarrow x = \pm \sqrt{2}.$$

We can exclude the negative solution, since $x_n > 0 \forall n$.

We now show that the limit exists. We first show that (x_n) is decreasing. We can calculate

$$x_{n+1} - x_n = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) - x_n = \frac{1}{x_n} - \frac{1}{2} x_n = \frac{2 - x_n^2}{2x_n}.$$

We thus need to show that $x_n^2 \geq 2 \forall n$. We do this by induction. The base case $n = 0$ is clear. Assume $x_n \geq \sqrt{2}$. Then

$$x_{n+1} - \sqrt{2} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) - \sqrt{2}.$$

Using the induction hypothesis, we have

$$= \frac{x_n^2 - 2\sqrt{2}x_n + 2}{2x_n} = \frac{(x_n - \sqrt{2})^2}{2x_n} \geq 0.$$

In general, we only care about the sequence for large n . This gives rise to the following definition.

Definition 2.96: Tail of a sequence

Given $(x_n)_{n=0}^\infty$ and $N \in \mathbb{N}$, the **TAIL** of the sequence is the sequence $X_{\geq n} = \{x_n, x_{n+1}, \dots\}$

Remark 2.97:

We can restate the definition of limit using tails of sequences:

$$\lim_{n \rightarrow \infty} x_n = A \Leftrightarrow \forall \varepsilon > 0 \exists N : X_{\geq N} \subseteq (A - \varepsilon, A + \varepsilon).$$

For simplicity we use the following notation:

$$\sup_{k \geq n} x_k \text{ for } \sup(X_{\geq n}) \text{ and } \inf_{k \geq n} x_k \text{ for } \inf(X_{\geq n}).$$

Let $(x_n)_{n \geq 0}$ be a bounded sequence, we define

$$s_n = \sup_{k \geq n} x_k \text{ and } i_n = \inf_{k \geq n} x_k.$$

If $m > n$, means that $X_{\geq m} \subseteq X_{\geq n}$, as we throw away more stuff. This implies that $s_m \leq s_n$ and $i_m \geq i_n$. So we have

$$i_n \leq i_m \leq s_m \leq s_n, \quad \forall m > n.$$

Thus, $(s_n)_{n \geq 0}$ is decreasing, $(i_n)_{n \geq 0}$ is increasing. since (x_n) is bounded, both (s_n) and (i_n) are bounded, this motivates to the following definition.

Definition 2.98: Superior and Inferior Limit

Let $(x_n)_{n \geq 0}$ be a bounded sequence. The numbers

$$\begin{aligned} \limsup_{n \rightarrow \infty} x_n &:= \lim_{n \rightarrow \infty} \underbrace{\left(\sup_{k \geq n} x_k \right)}_{=s_n} \\ \liminf_{n \rightarrow \infty} x_n &:= \lim_{n \rightarrow \infty} \underbrace{\left(\inf_{k \geq n} x_k \right)}_{=i_n} \end{aligned}$$

are called **SUPERIOR LIMIT** and **INFERIOR LIMIT**.

Also because $i_n \leq s_n$ it holds that

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

Lemma 2.99:

A bounded sequence $(x_n)_{n \geq 0}$ converges $\Leftrightarrow$

$$\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n.$$

Proof. We use the notation of i_n and s_n from before. Denote

$$I = \liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} i_n \text{ and } S = \limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} s_n.$$

$\Leftarrow$: Assume $I = S$. Note that $i_n \leq x_n \leq s_n \forall n \geq 0$. Thus, the sandwich lemma yields, $I = S \Rightarrow \lim x_n = I = S$.

$\Rightarrow$: Let $\lim_{n \rightarrow \infty} x_n = A$. Then, given $\varepsilon > 0, \exists N \in \mathbb{N}$ such that

$$A - \varepsilon < x_n < A + \varepsilon, \quad \forall n \geq N.$$

Therefore, for $n \geq N$,

$$A - \varepsilon \leq i_n \leq s_n \leq A + \varepsilon.$$

It follows that $A - \varepsilon \leq I \leq S \leq A + \varepsilon$. Because this is true for all $\varepsilon > 0$, we have $I = S = A$. $\square$

Theorem 2.100:

$(x_n)_{n \geq 0}$ be a bounded sequence, $A = \limsup_{n \rightarrow \infty} x_n$. Then A is an accumulation point of (x_n) .

Furthermore, $\forall \varepsilon > 0$,

- (1) only finitely many elements x_n satisfy

$$x_n \geq A + \varepsilon.$$

- (2) infinitely many elements satisfy

$$A - \varepsilon < x_n < A + \varepsilon.$$

Proof. Let $s_n = \sup_{k \geq n} x_k$. Then $A = \lim_{n \rightarrow \infty} s_n$, furthermore, (s_n) is decreasing.

Then, $\forall \varepsilon > 0 \exists N_0 \in \mathbb{N}$ such that

$$A \leq s_n < A + \varepsilon, \quad \forall n \geq N_0.$$

We want to prove, that A is an accumulation point. Fix $\varepsilon > 0$, fix $N \in \mathbb{N}$. take N_0 as before and define $N_1 = \max\{N, N_0\}$.

Since $s_{N_1} = \sup_{k \geq N_1} x_k$, $\exists n_1 \geq N_1$ such that

$$s_{N_1} - \varepsilon < x_{n_1} \leq s_{N_1}.$$

Then, $A - \varepsilon \leq s_{N_1} - \varepsilon < x_{n_1} \leq s_{N_1} < A + \varepsilon$.

This shows, that A is an accumulation point.

From the definition of N_0 , we have

$$x_n \leq s_n < A + \varepsilon, \quad \forall n \geq N_0.$$

Thus only finitely many elements can be $\geq A + \varepsilon$ (at most N_0).

Finally, because A is an accumulation point, there exists a subsequence $(x_{n_k})_{k=0}^\infty$ such that $\lim_{k \rightarrow \infty} x_{n_k} = A$. Therefore, $\forall \varepsilon > 0$, $\exists N_2 \in \mathbb{N}$ such that

$$A - \varepsilon < x_{n_k} < A + \varepsilon, \quad \forall k \geq N_2.$$

□

Corollary 2.101:

Every bounded sequence has at least one accumulation point and therefore possesses a convergent subsequence.

Proof. $(x_n)_{n \geq 0}$ bounded $\Rightarrow A = \limsup_{n \rightarrow \infty} x_n$ exists and is an accumulation point. Furthermore, every accumulation point is the limit of a subsequence.

□

Definition 2.102: Cauchy sequence

A sequence $(x_n)_{n \geq 0}$ is a **CAUCHY SEQUENCE** if

$$\forall \varepsilon > 0, \exists N : |x_n - x_m| < \varepsilon, \quad \forall n, m \geq N.$$

Lemma 2.103:

$(x_n)_{n \geq 0}$ **cauchy** $\Rightarrow (x_n)$ is bounded.

Proof. Fix $\varepsilon = 1$. Then $\exists N$ such that

$$|x_n - x_m| < 1, \quad \forall n, m \geq N.$$

Therefore, choosing $m = N$, we have

$$|x_n - x_N| < 1, \quad \forall n \geq N.$$

By triangle inequality, it follows that

$$|x_n| = |x_n - x_N + x_N| \leq |x_n - x_N| + |x_N| < 1 + |x_N|.$$

Choose $M = \max\{|x_0|, |x_1|, \dots, |x_{N-1}|, 1 + |x_N|\}$. Thus

$$|x_n| \leq M, \quad \forall n \in \mathbb{N}.$$

□

Theorem 2.104:

$(x_n)_{n \geq 0}$ is **cauchy** $\Leftrightarrow (x_n)$ is convergent.

Proof. $\Leftarrow$: Assume $\lim_{n \rightarrow \infty} x_n = A$. Fix $\varepsilon > 0$. Since x_n converges to A , $\exists N$, such that

$$|x_n - A| < \frac{\varepsilon}{2}, \quad \forall n \geq N.$$

Therefore, $\forall n, m \geq N$,

$$|x_n - x_m| = |x_n - A + A - x_m| \leq |x_n - A| + |x_m - A| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This proves that convergent series are **cauchy**.

$\Rightarrow$: If (x_n) is **cauchy**, we know that (x_n) is bounded, this implies that

$$\exists (x_{n_k})_{k \geq 0} : \lim_{k \rightarrow \infty} x_{n_k} = A.$$

Since (x_n) is **cauchy**, $\exists N_0$ such that

$$|x_n - x_m| < \frac{\varepsilon}{2}, \quad \forall n, m \geq N_0.$$

Since $\lim_{k \rightarrow \infty} x_{n_k} = A$, $\exists N_1$ such that

$$|x_{n_k} - A| < \frac{\varepsilon}{2}, \quad \forall k \geq N_1.$$

Let $N = \max\{N_0, n_{N_1}\}$. Then, $\forall n \geq N$,

$$|x_n - A| \leq |x_n - x_{n_N}| + |x_{n_N} - A| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

It is important to note that we need to have the inequality in the definition for all pairs, not just for consecutive terms.

Definition 2.105: Divergence

$(x_n)_{n \geq 0}$ **DIVERGES** to $+\infty$ if

$$\forall M > 0 \exists N : x_n > M, \quad \forall n \geq N.$$

Similarly, (x_n) **DIVERGES** to $-\infty$ if

$$\forall M > 0 \exists N : x_n < -M, \quad \forall n \geq N.$$

The sequence $0, 1, -2, 3, -4, 5, -6, \dots$ does not diverge nor converge by our definition.

Let (x_n) be a sequence, also unbounded. We define

$$\limsup_{n \rightarrow \infty} x_n = \begin{cases} \text{finite number} & \text{if } (x_n) \text{ is bounded} \\ +\infty & \text{if } (x_n) \text{ is not bdd above} \end{cases}.$$

2.5 Complex sequences

A sequence of complex numbers is the same as before, but now each number is complex:

$$z_n = x_n + iy_n, \quad x_n, y_n \in \mathbb{R}.$$

Definition 2.106: Convergence

A complex sequence $(z_n)_{n \geq 0}$ converges if

$$(x_n)_{n \geq 0}, (y_n)_{n \geq 0} \text{ both converge.}$$

In particular, if $\lim x_n = A$ and $\lim y_n = B$, then

$$\lim_{n \rightarrow \infty} z_n = A + iB.$$

We say that (z_n) **diverges** if $\{|z_n|\}_{n \geq 0}$ **diverges** to $+\infty$.

Most ideas from real sequences carry over to complex sequences. For example, a subsequence of (z_n) is (z_{n_k}) .

3 Functions of one Real Variable

3.1 Real Valued Functions

In the following we work with functions $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$.

For two functions f_1, f_2 we define

- $(f_1 + f_2)(x) = f_1(x) + f_2(x)$,
- $(\alpha f_1)(x) = \alpha f_1(x)$,
- $(f_1 \cdot f_2)(x) = f_1(x) \cdot f_2(x)$,

We call a point $x \in D$ a **ZERO** of f if $f(x) = 0$. The **ZERO SET** of f is $\{x \in D : f(x) = 0\}$.

We say that $f_1 \leq f_2$ if $f_1(x) \leq f_2(x) \forall x \in D$. Similarly we define $f_1 < f_2$. This defines an order relation.

We call f **NON-NEGATIVE** if $f \geq 0$ and **POSITIVE** if $f > 0$.

Definition 3.1: Bounded Functions

$f : D \rightarrow \mathbb{R}$ is **BOUNDED FROM ABOVE** if $\exists M > 0$ such that

$$f(x) \leq M, \quad \forall x \in D.$$

Similarly, f is **BOUNDED FROM BELOW** if $\exists M > 0$ such that

$$f(x) \geq -M, \quad \forall x \in D.$$

Finally, f is **BOUNDED** if

$$|f(x)| \leq M, \quad \forall x \in D.$$

Similar to sequences, we can define monotonicity for functions.

Definition 3.2: Monotonicity

Consider $f : D \rightarrow \mathbb{R}$ and $x, y \in D$.

- f is **INCREASING** if $x < y \Rightarrow f(x) \leq f(y)$.
- f is **STRICTLY INCREASING** if $x < y \Rightarrow f(x) < f(y)$.
- f is **DECREASING** if $x < y \Rightarrow f(x) \geq f(y)$.
- f is **STRICTLY DECREASING** if $x < y \Rightarrow f(x) > f(y)$.

Example 3.3: Parabola

$f(x) = x^2$ is strictly increasing on $D = [0, 1]$. However, it is strictly decreasing on $D = [-1, 0]$. Similarly, it is neither increasing nor decreasing on $D = [-1, 1]$.

We notice that the notion of monotonicity only makes sense when the domain is specified.

Example 3.4: Floor Function

The integer part (or floor) function is increasing on the domain $\mathbb{R}$, yet not strictly increasing.

If we have a constant function, we also write $f(x) \equiv c$. This is an example of a increasing and decreasing function.

Definition 3.5: Continuity

$D \subseteq \mathbb{R}$, $f : D \rightarrow \mathbb{R}$. We call f **CONTINUOUS** at $x_0 \in D$ if $\forall \varepsilon > 0, \exists \delta > 0$ such that

$$\forall x \in D : |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

f is called **CONTINUOUS** if it is continuous at every point $x_0 \in D$.

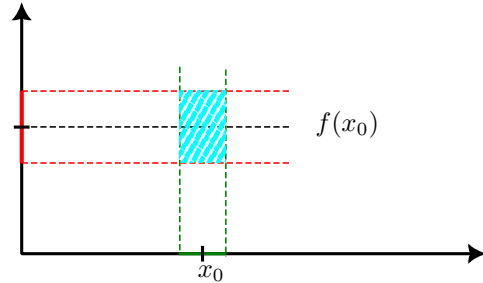


Figure 15: Epsilon-Delta definition of continuity.

Remark 3.6:

It is enough to consider ε small. More precisely, assume that $\exists \varepsilon_0$ such that $\forall \varepsilon \in (0, \varepsilon_0] \exists \delta$ such that

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

Then if $\varepsilon > \varepsilon_0$, just take for δ as the one for ε_0 .

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon_0 < \varepsilon.$$

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Example 3.7: Linear Functions

$f(x) = ax + b$. Show that f is continuous for all $x_0 \in \mathbb{R}$.

Solution. Fix x_0 . If $a = 0$ then $f(x)$ is constant which is continuous. Indeed, given $\varepsilon > 0$, just choose $\delta = 1$. Then

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| = |b - b| = 0 < \varepsilon.$$

If $a \neq 0$, given $\varepsilon > 0$, we want that

$$|x - x_0| < \delta \Rightarrow |ax + b - (ax_0 + b)| = |a||x - x_0| < \varepsilon.$$

Choose $\delta = \frac{\varepsilon}{|a|} > 0$. Then

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < |a||x - x_0| < |a|\delta = \varepsilon.$$

Example 3.8: Absolute Value Function

$x \rightarrow |x|$ is continuous for all $x_0 \in \mathbb{R}$.

Solution. Given $\varepsilon > 0$, we want to find $\delta > 0$ such that

$$|x - x_0| < \delta \Rightarrow ||x| - |x_0|| < \varepsilon.$$

Choose $\delta = \varepsilon$. Then, by the reverse triangle inequality,

$$|x - x_0| < \delta \Rightarrow ||x| - |x_0|| \leq |x - x_0| < \delta = \varepsilon.$$

Example 3.9: Integer part function

The function $f(x) = \lfloor x \rfloor$ is not continuous at the integers. This can be seen for example as $\lfloor 3 \rfloor = 3$ but

$$\lfloor 3 - \delta \rfloor = 2, \quad \forall \delta \in (0, 1).$$

Thus,

$$|f(3) - f(3 - \delta)| = |3 - 2| = 1 \forall \delta \in (0, 1).$$

Exercise 3.10:

Show that $f(x) = x^2$ is continuous for all $x_0 \in \mathbb{R}$ using the $\varepsilon - \delta$ definition.

Note: $\delta = \delta(\varepsilon, x_0)$.

Definition 3.11: Restriction

Given $f : D \rightarrow \mathbb{R}$, $D' \subseteq D$. Then $f|_{D'} : D' \rightarrow \mathbb{R}$ is the **RESTRICTION** of f to D' where $f|_{D'}(x) = f(x)$ for all $x \in D'$.

We look at f and $f|_{D'}$ as different functions.

For example, $\lfloor x \rfloor$ is not continuous on $\mathbb{R}$, but it is continuous on $D' = (0, 1)$. It is also continuous on $D'' = (1, 2) \cup (2, 3)$.

Proposition 3.12: Sum and Products of Continuous Functions

Given $f_1, f_2 : D \rightarrow \mathbb{R}$ both being continuous at $x_0 \in D$. Then:

1. $f_1 + f_2$ is continuous at x_0 .
2. $f_1 \cdot f_2$ is continuous at x_0 .
3. af_1 is continuous at x_0 for all $a \in \mathbb{R}$.

This implies that the set of continuous functions $C^0(D)$ is a vector space.

Proof. 1. Fix $\varepsilon > 0$. Since f_1, f_2 are continuous at x_0 , $\exists \delta_1, \delta_2 > 0$ such that

$$\begin{aligned} x \in D, |x - x_0| < \delta_1 &\Rightarrow |f_1(x) - f_1(x_0)| < \frac{\varepsilon}{2} \\ x \in D, |x - x_0| < \delta_2 &\Rightarrow |f_2(x) - f_2(x_0)| < \frac{\varepsilon}{2} \end{aligned}$$

If now we choose $\delta = \min\{\delta_1, \delta_2\}$, then

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| \leq |f_1(x) - f_1(x_0)| + |f_2(x) - f_2(x_0)| < \varepsilon.$$

2. We want to estimate $|f_1(x)f_2(x) - f_1(x_0)f_2(x_0)|$. We rewrite this as

$$|f_1(x)f_2(x) - f_1(x_0)f_2(x) + f_1(x_0)f_2(x) - f_1(x_0)f_2(x_0)|.$$

Applying triangle inequality, we have

$$\begin{aligned} |f_1(x)f_2(x) - f_1(x_0)f_2(x_0)| \\ \leq |f_1(x) - f_1(x_0)||f_2(x)| + |f_2(x) - f_2(x_0)||f_1(x_0)| \end{aligned}$$

We apply continuity to f_2 with $\varepsilon = 1$. Thus $\exists \delta_0 > 0$ such that

$$|x - x_0| < \delta_0 \Rightarrow |f_2(x) - f_2(x_0)| < 1 \Rightarrow |f_2(x)| < 1 + |f_2(x_0)|.$$

Apply again continuity to f_1, f_2

$$\begin{aligned} \exists \delta_1 : |x - x_0| < \delta_1 &\Rightarrow |f_1(x) - f_1(x_0)| < \frac{\varepsilon}{2(1 + |f_2(x_0)|)} \\ \exists \delta_2 : |x - x_0| < \delta_2 &\Rightarrow |f_2(x) - f_2(x_0)| < \frac{\varepsilon}{2(1 + |f_1(x_0)|)}. \end{aligned}$$

For the second inequality we use $+1$ in the denominator to avoid division by zero. Finally, choose $\delta = \min\{\delta_0, \delta_1, \delta_2\}$. Then, $\forall x \in D$ with $|x - x_0| < \delta$,

$$|f_1(x)f_2(x) - f_1(x_0)f_2(x_0)| < \frac{\varepsilon \cdot |f_2(x)|}{2(1 + |f_2(x_0)|)} + \frac{\varepsilon \cdot |f_1(x_0)|}{2(1 + |f_1(x_0)|)} < \varepsilon.$$

3. Apply the second point to $f_2 \equiv a$. □

We use the following notation:

$$\sum_{j=0}^N a_j = a_0 + a_1 + \cdots + a_N.$$

Remark 3.13: Polynomials are continuous

Given a **POLYNOMIAL** $f(x) = \sum_{j=0}^N a_j x^j$, where $x^j = \underbrace{x \cdot x \cdots x}_{j \text{ times}}$. Then f is a continuous function.

Given a function $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, we can $g \circ f : X \rightarrow Z$, where $(g \circ f)(x) = g(f(x))$. This is called the **COMPOSITION** of f and g .

Proposition 3.14: Continuity of Composition

Given $f : D_1 \rightarrow D_2$ continuous at x_0 , $g : D_2 \rightarrow \mathbb{R}$ continuous at $f(x_0)$. Then $g \circ f : D_1 \rightarrow \mathbb{R}$ is continuous at x_0 .

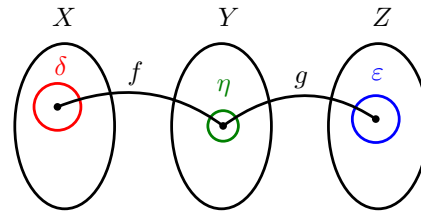


Figure 16: Composition of functions.

Proof. Fix $\varepsilon > 0$. Since g is continuous at $f(x_0)$, $\exists \eta > 0$ such that

$$y \in D_2, |y - f(x_0)| < \eta \Rightarrow |g(y) - g(f(x_0))| < \varepsilon.$$

Since f is continuous at x_0 , $\exists \delta > 0$ such that

$$x \in D_1, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \eta.$$

Combining both, we have

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \eta \Rightarrow |g(f(x)) - g(f(x_0))| < \varepsilon.$$

□

Exercise 3.15:

Show the following:

1. $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x}$ is continuous,
2. $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) \neq 0$ is continuous, then $\frac{1}{g(x)}$ is continuous,
3. $g, h : \mathbb{R} \rightarrow \mathbb{R}$ continuous, $g(x) \neq 0$, then $\frac{h(x)}{g(x)}$ is continuous.

As a notation, instead of writing $\lim_{n \rightarrow \infty} x_n = \bar{x}$, we can write $x_n \rightarrow \bar{x}$.

Theorem 3.16: Continuity = Sequential Continuity

$f : D \rightarrow \mathbb{R}, x \in D$. f is continuous at $\bar{x} \Leftrightarrow$

$$\forall (x_n)_{n \geq 0} \subseteq D, \text{ if } x_n \rightarrow \bar{x} \text{ then } f(x_n) \rightarrow f(\bar{x}).$$

Proof. $\Rightarrow$: Given $\varepsilon > 0 \exists \delta > 0$ such that

$$|x - \bar{x}| < \delta \Rightarrow |f(x) - f(\bar{x})| < \varepsilon.$$

Since $x_n \rightarrow \bar{x}$, $\exists N$ such that

$$|x_n - \bar{x}| < \delta, \quad \forall n \geq N.$$

Then,

$$\forall n \geq N : |f(x_n) - f(\bar{x})| < \varepsilon.$$

So $f(x_n) \rightarrow f(\bar{x})$.

$\Leftarrow$: Assume that f is not continuous at $\bar{x}$. Then $\exists \varepsilon_0 > 0$ such that:

$$\forall \delta > 0, \exists x : |x - \bar{x}| < \delta \text{ but } |f(x) - f(\bar{x})| \geq \varepsilon_0.$$

Apply this with $\delta = 2^{-n}$ to find x_n such that

$$|x_n - \bar{x}| < 2^{-n} \text{ but } |f(x_n) - f(\bar{x})| \geq \varepsilon_0.$$

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Then $x_n \rightarrow \bar{x}$ but $f(x_n) \not\rightarrow f(\bar{x})$. $\square$

Remark 3.17:

If f is not continuous at $\bar{x} \exists \varepsilon > 0$ and $(x_n)_{n \geq 0}$ such that

$$x_n \rightarrow \bar{x} \text{ but } |f(x_n) - f(\bar{x})| \geq \varepsilon.$$

3.2 Continuous Functions

The following theorem is what basically guarantees that continuous functions can be drawn without lifting the pen.

Theorem 3.18: Intermediate Value Theorem

Given $f : [a, b] \rightarrow \mathbb{R}$ continuous. Assume $f(a) < f(b)$. Then

$$\forall c \in [f(a), f(b)] \exists \bar{x} \in [a, b] : f(\bar{x}) = c.$$

Proof. Define $X = \{x \in [a, b] \mid f(x) \leq c\}$. Then X is bounded since $X \subseteq [a, b]$. Also, $X \neq \emptyset$ since $a \in X$. Thus, $\exists \bar{x} = \sup(X)$.

The goal is to show that $f(\bar{x}) = c$.

1. By definition of the supremum, $\forall n \in \mathbb{N} \exists x_n \in [\bar{x} - 2^{-n}, \bar{x}]$ such that $f(x_n) \leq c$. This follows as the supremum is the least upper bound.

Since $x_n \rightarrow \bar{x}$, by continuity of f , $f(x_n) \rightarrow f(\bar{x})$. Since all $f(x_n) \leq c$, we have $f(\bar{x}) \leq c$.

2. Assume by contradiction that $f(\bar{x}) < c$. Define $\varepsilon = c - f(\bar{x}) > 0$. By continuity, $\exists \delta > 0$ such that

$$|x - \bar{x}| < \delta \Rightarrow |f(x) - f(\bar{x})| < \varepsilon.$$

As $|a| < b \Rightarrow -b < a < b$, we have

$$f(x) < f(\bar{x}) + \varepsilon = c.$$

Consider the two cases:

Case 1: $c = f(b)$. In that case, just take $\bar{x} = b$.

Case 2: $c < f(b)$. Then since $f(\bar{x}) \leq c$, then $\bar{x} < b$. But then $(\bar{x}, \bar{x} + \delta) \cap (\bar{x}, b) \neq \emptyset$. and there, as we have shown, $f(x) < c$. But this means

$$(\bar{x}, \bar{x} + \delta) \cap (\bar{x}, b) \subseteq X.$$

This shows, that there is a point to the right of $\bar{x}$ in X . However, this contradicts the fact that $\bar{x} = \sup(X)$. $\square$

Remark 3.19:

The same holds, if $f(a) \geq f(b)$, with $c \in [f(b), f(a)]$.

Claim 3.20:

Given $f : [a, b] \rightarrow [a, b]$ continuous. Then f has a fixpoint i.e. $f(\bar{x}) = \bar{x}$ for some $\bar{x} \in [a, b]$.

Proof. Let $h(x) = f(x) - x$. Notice that h is continuous as it is the sum of two continuous functions. Also, zeros of h are fixpoints of f . We distinguish three cases:

1. If $f(a) = a$ then a is a fixpoint.

2. If $f(b) = b$ then b is a fixpoint.

3. If $f(a) > a$ and $f(b) < b$, then $h(a) > 0$ and $h(b) < 0$. By the intermediate value theorem, $\exists \bar{x} \in [a, b]$ such that $h(\bar{x}) = 0$. Thus, $f(\bar{x}) = \bar{x}$. $\square$

Example 3.21:

Given $f : \mathbb{R} \rightarrow \mathbb{R}$ continuous and periodic with period 4, i.e.

$$\forall x \in \mathbb{R}, f(x + 4) = f(x).$$

Show that $\exists y \in [0, 2]$ such that $f(y) = f(y + 2)$.

Solution. Let $h(x) = f(x) - f(x + 2)$. Then h is continuous as it is the sum of two continuous functions. Looking at $h(0)$ and $h(2)$, we see

$$h(0) = f(0) - f(2)$$

$$h(2) = f(2) - f(4) = f(2) - f(0) = -h(0).$$

We have two cases: 1. If $h(0) = h(2) = 0$, then $f(0) = f(2)$ and we are done.

2. Either $h(0) > 0$ and thus $h(2) < 0$ or $h(0) < 0$ and thus $h(2) > 0$. In both cases, by the intermediate value theorem, $\exists y \in [0, 2]$ such that $h(y) = 0$. Thus, $f(y) = f(y + 2)$.

Tip 3.22:

To apply the intermediate value theorem, it is often useful to define a new function as the difference of two functions.

Definition 3.23: Inverse Function

Given: $f : X \rightarrow Y$ bijective, $\forall y \in Y \exists! x \in X : f(x) = y$. Then define $g(y) = x$. Then $g : Y \rightarrow X$ and

$$g \circ f : X \rightarrow X, g \circ f = \text{id}_X, f \circ g : Y \rightarrow Y, f \circ g = \text{id}_Y.$$

g is called the **INVERSE** of f and is denoted by f^{-1} .

Theorem 3.24: Inverse Function Theorem

Given $f : I \rightarrow \mathbb{R}$, where $I \subseteq \mathbb{R}$ is an interval, such that f is continuous and strictly monotone.

Then $f(I)$ is an interval and $f : I \rightarrow f(I)$ has a continuous, strictly monotone inverse $f^{-1} : f(I) \rightarrow I$.

Proof. We can assume f is strictly increasing (otherwise, consider $-f$). Let $J = f(I)$.

- f strictly increasing ($x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$) $\Rightarrow f$ is injective.

- So $f : I \rightarrow J$ is bijective by the definition of the image. Thus, $\exists g = f^{-1} : J \rightarrow I$ which is the inverse of f .

1. g is strictly increasing. Indeed,

$$x_1 < x_2 \Leftrightarrow \underbrace{f(x_1)}_{y_1} < \underbrace{f(x_2)}_{y_2} \Leftrightarrow g(y_1) < g(y_2).$$

2. For J to be an interval, $\forall y_1, y_2 \in J, y_1 < y_2$ implies $[y_1, y_2] \subseteq J$.

Fix $y_1 < y_2 \in J$. Then $\exists x_1 < x_2 : f(x_1) = y_1, f(x_2) = y_2$. Let $c \in [y_1, y_2]$. By the intermediate value theorem applied to $f : [x_1, x_2] \rightarrow \mathbb{R}$, $\exists \bar{x} \in [x_1, x_2]$ such that $f(\bar{x}) = c$. Thus, $c \in J$. So $[y_1, y_2] \subseteq J$.

This proves that J is an interval.

3. By contradiction, assume that $g = f^{-1}$ is not continuous at some point $\bar{y} \in J$. By the remark 3.17, $\exists \varepsilon > 0$ and $(y_n)_{n \geq 0} \subseteq J$ such that

$$y_n \rightarrow \bar{y} \text{ but } |g(y_n) - g(\bar{y})| \geq \varepsilon.$$

Let $x_n = g(y_n)$, $\bar{x} = g(\bar{y})$. Then $y_n = f(x_n) \rightarrow \bar{y} = f(\bar{x})$ and $|x_n - \bar{x}| \geq \varepsilon$.

Assume that there are infinitely many x_n such that $x_n \leq \bar{x} - \varepsilon$. Considering these elements, we get a subsequence $(x_{n_k})_{k \geq 0}$ such that

$$f(x_{n_k}) \rightarrow f(\bar{x}) \text{ but } x_{n_k} \leq \bar{x} - \varepsilon.$$

Thus we have by strict monotonicity

$$f(\bar{x}) < -f(x_{n_k}) \leq f(\bar{x} - \varepsilon) < f(\bar{x}).$$

But then $f(\bar{x}) < f(\bar{x})$, which is a contradiction. $\square$

Remark 3.25:

We know that $x \mapsto x^n$ is continuous and on $[0, \infty) \mapsto [0, \infty)$, they are strictly increasing. Thus $\exists$ inverse

$$\begin{aligned} \sqrt[n]{\cdot} : [0, \infty) &\longrightarrow [0, \infty) \\ y &\longmapsto \sqrt[n]{y}. \end{aligned}$$

Furthermore, $x \mapsto x^{\frac{m}{n}} = \underbrace{\sqrt[n]{x} \cdots \sqrt[n]{x}}_{m \text{ times}}$ is continuous on $[0, \infty)$.

Also for $x > 0$, $x^{-\frac{m}{n}} = \frac{1}{x^{\frac{m}{n}}}$ is continuous.

To draw an inverse of a function, we can reflect the graph of the function along the line $y = x$.

Example 3.26:

Show that $f(x) = (x-5)(x-3)(x-2)(x+1)(x+2)+1$ has 5 zeros.

Solution. Since f is a polynomial, it is continuous. Also, note that for example $f(-5) < 0$, $f(-1.5) > 0$ and $f(-0.5) < 0$. Thus by intermediate value theorem, $\exists x_1 \in [-5, -1.5]$ such that $f(x_1) = 0$ and $\exists x_2 \in [-1.5, -0.5]$ such that $f(x_2) = 0$. Similarly, we can find the existence of 3 more zeros.

3.3 Continuous Functions on Compact Intervals

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We like to show some properties of continuous functions on compact intervals.

Definition 3.27: Compact Interval

A bounded, closed interval $[a, b]$ is called **COMPACT**.

For example, $[a, \infty]$ is closed yet not bounded, so not compact.

Lemma 3.28:

Let $(x_n)_{n \geq 0} \subseteq [a, b]$ be a sequence. Then $\exists$ a subsequence $(x_{n_k})_{k \geq 0}$ such that $x_{n_k} \rightarrow \bar{x} \in [a, b]$.

Proof. Since $[a, b]$ is bounded, this implies that $(x_n)_{n \geq 0}$ is bounded. Thus, $\exists (x_{n_k})_{k \geq 0}$ converges to some $\bar{x} \in \mathbb{R}$, because every bounded sequence has a convergent subsequence (lim sup for example). Furthermore, because $a \leq x_{n_k} \leq b$, we have $a \leq \bar{x} \leq b$. $\square$

Tip 3.29:

When working with compact intervals, the typical strategy is to proof by contradiction and then find a convergent subsequence.

Theorem 3.30:

Given $f : [a, b] \rightarrow \mathbb{R}$ continuous. Then f is bounded.

Proof. By contradiction, assume f is not bounded. Then $\forall n \in \mathbb{N}, \exists x_n \in [a, b] : |f(x_n)| \geq n$.

By the lemma, $\exists (x_{n_k})_{k \geq 0}$ such that $x_{n_k} \rightarrow \bar{x} \in [a, b]$.

$\square$ So $|f(x_{n_k})| \geq n_k \rightarrow \infty$. But since f is continuous, also $|f|$ is continuous. Then $f(x_{n_k}) \rightarrow f(\bar{x})$, which is a real number. ζ $\square$

Definition 3.31: Maximum & Minimum Value

Given $f : D \rightarrow \mathbb{R}$. We say that f takes its **MAXIMUM VALUE** at x_0 if

$$f(x) \leq f(x_0) \quad \forall x \in D.$$

We call $f(x_0)$ the **MAXIMUM** of f .

The same for **MINIMUM VALUE** and **MINIMUM**.

Theorem 3.32:

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then f takes its maximum and minimum value.

Proof. We just proof it for the maximum. Since f is bounded, $f([a, b])$ is bounded. Thus, $\exists S = \sup(f([a, b]))$.

$\forall n \in \mathbb{N}, \exists y_n \in f([a, b])$ such that $S - 2^{-n} \leq y_n \leq S$. Since y_n is in the image of f , $\exists x_n \in [a, b]$ such that $f(x_n) = y_n$.

Let $x_{n_k} \rightarrow \bar{x} \in [a, b]$ be a convergent subsequence. Since $S - 2^{-n_k} \leq f(x_{n_k}) = y_{n_k} \leq S$. Since f is continuous, we get $f(x_{n_k}) \rightarrow f(\bar{x})$. Thus,

$$S \leq f(\bar{x}) \leq S \Rightarrow f(\bar{x}) = S.$$

$\square$

For all of the above properties, it is crucial that the interval is compact. As a conterexample, consider $f : (0, 1) \rightarrow \mathbb{R}, x \mapsto \frac{1}{x}$.

We recall the definition of a continuous function on a set $D \subseteq \mathbb{R}$:

$$\forall x_0 \in D, \forall \varepsilon > 0, \exists \delta > 0 : |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

A diffrent thing is uniform continuity:

Definition 3.33: Uniform Continuity

$f : D \rightarrow \mathbb{R}$ is **UNIFORMLY CONTINUOUS** if $\forall \varepsilon > 0, \exists \delta > 0$ such that

$$\forall x, y \in D, |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

The difference to regular continuity is that δ does not depend on x_0 .

Example 3.34:

Consider $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$. Then f is continuous but not uniformly continuous.

Solution. Indeed, take $\varepsilon = \frac{1}{2}$ for example. We have to show that $\forall \delta > 0, \exists x, y \in \mathbb{R}$ such that

$$|x - y| < \delta \text{ but } |f(x) - f(y)| \geq \varepsilon.$$

Now take x and $y = x + \frac{\delta}{2}$. Then, plugging in, we get $f(x) - f(y) = \delta x + \frac{\delta^2}{4}$. For $x = \frac{1}{\delta}$, we get

$$|f(x) - f(y)| = 1 + \frac{\delta^2}{4} > \frac{1}{2}.$$

Theorem 3.35: Heine-Cantor Theorem

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then f is uniformly continuous.

Proof. If f is not uniformly continuous, then $\exists \varepsilon > 0$ such that $\forall \delta > 0, \exists x, y \in [a, b]$ such that

$$|x - y| < \delta \text{ but } |f(x) - f(y)| \geq \varepsilon.$$

For $n \in \mathbb{N}$, apply with $\delta = 2^{-n}$. Then $\exists x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < 2^{-n} \text{ but } |f(x_n) - f(y_n)| \geq \varepsilon.$$

Let $(x_{n_k})_{k \geq 0}$ be a subsequence of $(x_n)_{n \geq 0}$ with $x_{n_k} \rightarrow \bar{x}$.

Note:

$$|y_{n_k} - \bar{x}| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - \bar{x}| < 2^{-n_k} + |x_{n_k} - \bar{x}| \xrightarrow{k \rightarrow \infty} 0.$$

Thus, also $y_{n_k} \rightarrow \bar{x}$. Since f is continuous, it follows that $f(x_{n_k}) \rightarrow f(\bar{x})$ and $f(y_{n_k}) \rightarrow f(\bar{x})$. Thus

$$\varepsilon \leq |f(x_{n_k}) - f(y_{n_k})| \leq |f(x_{n_k}) - f(\bar{x})| + |f(\bar{x}) - f(y_{n_k})| \rightarrow 0.$$

□

3.4 Exponential and Logarithmic Functions

We start with a lemma.

Lemma 3.36: Bernoulli inequality

$$\forall a \in \mathbb{R}, a \geq -1, n \geq 1 \Rightarrow (1 + a)^n \geq 1 + na.$$

Proof. We proof by induction. The base case $n = 1$ is clear. Assume the statement holds for some $n \geq 1$. Then

$$\begin{aligned} (1 + a)^{n+1} &= (1 + a)^n (1 + a) \\ &\geq (1 + na)(1 + a) \\ &= 1 + (n + 1)a + na^2 \\ &\geq 1 + (n + 1)a. \end{aligned}$$

□

The goal is to define $e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$.

Proposition 3.37:

Fix $x \in \mathbb{R}$. Then the sequence $(a_n)_{n \geq 1}$ where

$$a_n = \left(1 + \frac{x}{n}\right)^n$$

is convergent and its limit is positive.

Lemma 3.38:

Fix $x \in \mathbb{R}$, let $n_0 > -x$. Then the sequence $(a_n)_{n \geq n_0}$ is increasing.

Proof. We have

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{\left(1 + \frac{x}{n+1}\right)^{n+1}}{\left(1 + \frac{x}{n}\right)^n} \\ &= \left(1 + \frac{x}{n+1}\right)^n \cdot \frac{1 + \frac{x}{n+1}}{1 + \frac{x}{n}} \\ &= \left(\frac{n+x}{n}\right) \cdot \left(1 - \frac{x}{(n+1)(n+x)}\right)^{n+1} \end{aligned}$$

For $n \geq n_0$, we have $\frac{x}{(n+1)(n+x)} \leq \frac{x+n}{(n+1)(n+x)} < 1$.

This allows us to apply bernoulli's inequality:

$$\begin{aligned} \frac{a_{n+1}}{a_n} &\geq \left(\frac{n+x}{n}\right) \cdot \left(1 - (n+1) \frac{x}{(n+1)(n+x)}\right) \\ &= \left(\frac{n+x}{n}\right) \cdot \left(1 - \frac{x}{n+x}\right) \\ &= \left(\frac{n+x}{n}\right) \cdot \left(\frac{n}{n+x}\right) = 1. \end{aligned}$$

□

Proof. [Proposition] Fix $x \in \mathbb{R}$, $n_0 > -x$. We know, that $(a_n)_{n \geq n_0}$ is increasing. We need to prove that the sequence is bounded.

1. $x \leq 0$. For $n > n_0$, $0 < 1 + \frac{x}{n} \leq 1$. This implies $0 < a_n \leq 1$. Thus, $(a_n)_{n \geq n_0}$ converges to a positive limit.

2. $x > 0$. For $n > x$, $\left(1 + \frac{x}{n}\right)^n \left(1 - \frac{x}{n}\right)^n = \left(1 - \frac{x^2}{n^2}\right)^n \leq 1$. This means

$$\left(1 + \frac{x}{n}\right)^n \leq \frac{1}{\left(1 - \frac{x}{n}\right)^n}.$$

But the denominator, is exactly what we had in case 1. Thus, we know that $\left(1 - \frac{x}{n}\right)^n$ converges to a value in $(0, 1]$. Since this value is non-zero, we know that also $\frac{1}{\left(1 - \frac{x}{n}\right)^n}$ is bounded.

Thus, $(a_n)_{n \geq n_0}$ is bounded from above and increasing, so convergent.

□

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Definition 3.39: Exponential Function

Given $x \in \mathbb{R}$. We define the **EXPONENTIAL FUNCTION** as

$$\exp(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n.$$

This gives $\exp : \mathbb{R} \rightarrow \mathbb{R}_{>0}$.

Definition 3.40: Euler's Number

Euler's number e is defined as $e = \exp(1) \approx 2.71828$.

Corollary 3.41:

For $x \geq -n$, we have

$$\exp(x) \geq \left(1 + \frac{x}{n}\right)^n.$$

Proof. Since $(a_n)_{n \geq n_0}$ is increasing, we have

$$\exp(x) = \lim_{n \rightarrow \infty} a_n \geq a_n = \left(1 + \frac{x}{n}\right)^n.$$

□

Corollary 3.42:

If $x \geq 0$, then $\exp(x) \geq \frac{x^n}{n^n}$.

Corollary 3.43:

$\exp(x) \geq 1 + x$.

Proof. By corollary 3.41, $\exp(x) > 1 + x$ for $x > -1$. For $x \leq -1$, $1 + x \leq 0 < \exp(x)$. $\square$

Theorem 3.44: Properties of exp

For the exponential function, the following properties hold:

- $\exp(0) = 1$.
- $\exp(-x) = \frac{1}{\exp(x)}$.
- $\exp(x + y) = \exp(x) \cdot \exp(y)$.
- $\exp$ is continuous, strictly increasing and bijective from $\mathbb{R}$ to $\mathbb{R}_{>0}$.

Proof. 1. $\exp(0) = \lim_{n \rightarrow \infty} \left(1 + \frac{0}{n}\right)^n = 1$.

2.

$$\begin{aligned} \exp(x) \cdot \exp(-x) &= \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n \cdot \left(1 + \frac{-x}{n}\right)^n \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{x^2}{n^2}\right)^n \leq 1. \end{aligned}$$

By Bernoulli's inequality, we also have $1 - \frac{x^2}{n} \leq \left(1 - \frac{x^2}{n^2}\right)^n$. But as $n \rightarrow \infty$, $1 - \frac{x^2}{n} \rightarrow 1$. Thus, by sandwich lemma, the limit is 1.

3. $\exp(x + y) \exp(-x) \exp(-y) = \dots$. The proof is left as an exercise.

4. Continuity: Recall that $\exp(x) \geq 1 + x$. Therefore,

$$\exp(x) = \frac{1}{\exp(-x)} \leq \frac{1}{1 - x} \quad \text{for } x < 1.$$

Let $x \in (-\delta, \delta)$ for some $\delta > 0$. Then

i. If $x > 0$, $0 \leq \exp(x) - \exp(0) = \exp(x) - 1 \leq \frac{1}{1-x} - 1 = \frac{x}{1-x}$. This is then bounded by $\frac{\delta}{1-\delta}$.

ii. If $x < 0$, $0 \geq \exp(x) - \exp(0) = \exp(x) - 1 \geq x \geq -\delta$.

Therefore, since $\delta \leq \frac{\delta}{1-\delta}$,

$$\forall x \in (-\delta, \delta), |\exp(x) - \exp(0)| \leq \frac{\delta}{1-\delta}.$$

Given $\varepsilon > 0$, choose $\delta = \frac{\varepsilon}{1+\varepsilon}$. Then $\frac{\delta}{1-\delta} = \varepsilon$. Thus, $\exp$ is continuous at 0.

To show, $\exp$ is continuous at arbitrary $x \in \mathbb{R}$, note that

$$\exp(x) = \exp(x - x_0) \exp(x_0).$$

For strict monotonicity, let $y > x$. This implies that

$$\exp(y) = \exp(x) \exp(y - x) > \exp(x) \cdot 1 = \exp(x).$$

The proof for bijectivity can be found in the notes. $\square$

Definition 3.45:

We define the **LOGARITHMIC FUNCTION** $\log : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ as the inverse of the exponential function. It satisfies the following properties:

- $\log(1) = 0$.
- $\log(xy) = \log(x) + \log(y)$.
- $\log$ is continuous, strictly increasing and bijective from $\mathbb{R}_{>0}$ to $\mathbb{R}$.

Given $a > 1$, we define

$$\log_a(x) = \frac{\log(x)}{\log(a)}.$$

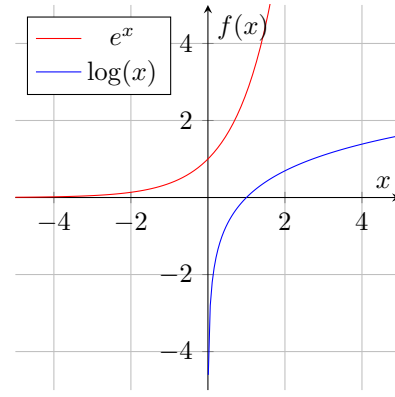


Figure 17: Exponential and Logarithmic Function

We also can define a^x

$$a^x = \exp(x \log(a)) \quad \text{and} \quad x^a = \exp(a \log(x)).$$

3.5 Limits of Functions

Consider the following definition.

Definition 3.46: Accumulation Points of Sets

Given $D \subseteq \mathbb{R}$, $x_0 \in \mathbb{R}$. x_0 is an **ACCUMULATION POINT** of D if

$$\forall \delta > 0, (x_0 - \delta, x_0 + \delta) \cap D \setminus \{x_0\} \neq \emptyset.$$

Definition 3.47: Limit of a Function

Given $f : D \rightarrow \mathbb{R}$, x_0 an accumulation point of D . We say that $L \in \mathbb{R}$ is the **LIMIT** of f as $x \rightarrow x_0$ if

$$\forall \varepsilon > 0, \exists \delta > 0 : x \in D, |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

We write $L = \lim_{x \rightarrow x_0} f(x)$.

This definition is very similar to the definition of continuity, however, f does not need to be defined at x_0 .

Remark 3.48:

If $\lim_{x \rightarrow x_0} f(x) = L_1$ and $\lim_{x \rightarrow x_0} g(x) = L_2$, then

- $\lim_{x \rightarrow x_0} (f(x) + g(x)) = L_1 + L_2$.
- $\lim_{x \rightarrow x_0} (f(x) \cdot g(x)) = L_1 \cdot L_2$.
- If $f \leq g$, then $L_1 \leq L_2$.
- If $f \leq h \leq g$ and $L_1 = L_2$, then $\lim_{x \rightarrow x_0} h(x) = L_1$.

Remark 3.49:

If $x_0 \in D$, then f is continuous at x_0 iff $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Let us assume that x_0 is an accumulation point for $D \setminus \{x_0\}$. Further assume that

$$L = \lim_{x \rightarrow x_0} f|_{D \setminus \{x_0\}}(x)$$

exists.

If $x_0 \in D$, assume $f(x_0) \neq L$. In this case, x_0 is called a **REMOVABLE DISCONTINUITY** of f .

Instead of $\lim_{x \rightarrow x_0} f|_{D \setminus \{x_0\}}(x)$, we also write

$$\lim_{\substack{x \rightarrow x_0 \\ x \neq x_0}} f(x) = L.$$

For x_0 to be a removable discontinuity, it is necessary that

$$\tilde{f} = \begin{cases} f(x) & x \neq x_0 \\ L & x = x_0 \end{cases}.$$

is continuous at x_0 .

If $x_0 \notin D$, then the function

$$\tilde{f} = \begin{cases} f(x) & x \in D \\ L & x = x_0 \end{cases}.$$

Is called the **CONTINUOUS EXTENSION** of f at x_0 .

Lemma 3.50:

$f : D \rightarrow \mathbb{R}$. Then $\lim_{x \rightarrow \bar{x}} f(x) = L$ iff

$$\forall (x_n)_{n \geq 0} \subseteq D, x_n \rightarrow \bar{x}, f(x_n) \rightarrow L.$$

Proposition 3.51:

$D, E \subseteq \mathbb{R}, f : D \rightarrow E$, assume that $L = \lim_{x \rightarrow \bar{x}} f(x)$ and $L \in E$.

Then, if $g : E \rightarrow \mathbb{R}$ with g continuous at L , we have

$$\lim_{x \rightarrow \bar{x}} g(f(x)) = g(L).$$

Proof. Let $(x_n) \subseteq D, x_n \rightarrow \bar{x}$. Then by previous lemma, $f(x_n) \rightarrow L$. Since g is continuous at L , we have

$$g(f(x_n)) \rightarrow g(L).$$

By the lemma again, this implies

$$g(f(x)) \xrightarrow{x \rightarrow \bar{x}} g(L).$$

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□

In the definition of the limit before, we required that $L \in \mathbb{R}$. However, we can extend this definition as follows.

Definition 3.52: Improper Limit

Let $f : D \rightarrow \mathbb{R}, x_0$ be an accumulation point of D . We say that f **DIVERGES** to $+\infty$ as $x \rightarrow x_0$ if

$$\forall M > 0, \exists \delta > 0 : x \in D, |x - x_0| < \delta \Rightarrow f(x) \geq M.$$

In this case, we write

$$\lim_{x \rightarrow x_0} f(x) = +\infty.$$

We say $\lim_{x \rightarrow x_0} f(x) = -\infty$ if

$$\forall M > 0, \exists \delta > 0 : x \in D, |x - x_0| < \delta \Rightarrow f(x) \leq -M.$$

Definition 3.53: One-Sided Limits

x_0 is an **ACCUMULATION POINT FROM THE RIGHT** if

$$(x_0, x_0 + \delta) \cap D \neq \emptyset, \quad \forall \delta > 0.$$

We say that $\lim_{x \rightarrow x_0^+} f(x) = L$ if

$$\forall \varepsilon > 0, \exists \delta > 0 : x \in D \cap (x_0, x_0 + \delta) \Rightarrow |f(x) - L| < \varepsilon.$$

Same from **LEFT**. In this case we write $\lim_{x \rightarrow x_0^-} f(x) = L$.

Essentially, an accumulation point of a set D is a point where we can construct a sequence in D that converges to that point.

Definition 3.54: Limits at Infinity

Let $f : D \rightarrow \mathbb{R}$. assume $D \cap (R, \infty) \neq \emptyset \forall R > 0$.

We say that $\lim_{x \rightarrow \infty} f(x) = L$ if

$$\forall \varepsilon > 0, \exists R > 0 : x \in D \cap (R, \infty) \Rightarrow |f(x) - L| < \varepsilon.$$

The limit at $-\infty$ is defined analogously.

If we define $g(x) = f(\frac{1}{x})$, then we can relate limits at infinity to limits at 0 with

$$\lim_{x \rightarrow \infty} f(x) = L \Leftrightarrow \lim_{x \rightarrow 0^+} g(x) = L.$$

Similarly for $-\infty$ and 0^- .

$$\lim_{x \rightarrow -\infty} f(x) = L \Leftrightarrow \lim_{x \rightarrow 0^-} g(x) = L.$$

Definition 3.55: One-Sided Continuity

Given $f : D \rightarrow \mathbb{R}, x_0 \in D$. Then f is **CONTINUOUS FROM THE RIGHT** at x_0 if

$$f(x_0) = \lim_{x \rightarrow x_0^+} f(x).$$

f is **CONTINUOUS FROM THE LEFT** at x_0 if

$$f(x_0) = \lim_{x \rightarrow x_0^-} f(x).$$

Example 3.56:

Let

$$f(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases}$$

This function is continuous from the right at 0, but not from the left.

If the limit from the right and left both exist and are different, then x_0 is a **JUMP POINT** of f .

Example 3.57:

$D = (0, \infty); f(x) = x^x$ Goal: Compute $\lim_{x \rightarrow x_0^+} f(x)$.

Solution. We perform the proof in 3 steps.

1. We compute $\lim_{y \rightarrow \infty} ye^{-y}$. Recall that $e^y \geq (1 + \frac{y}{n})^n$. In particular,

$$e^y \geq (1 + \frac{y}{2})^2 \Rightarrow e^{-y} \leq \frac{1}{(1 + \frac{y}{2})^2}.$$

Therefore,

$$0 \leq ye^{-y} \leq \frac{y}{(1 + \frac{y}{2})^2} \leq \frac{4}{y} \xrightarrow{y \rightarrow \infty} 0.$$

Thus by sandwich lemma, $\lim_{y \rightarrow \infty} ye^{-y} = 0$.

2. Compute the limit $\lim_{x \rightarrow 0^+} x \log(x)$. By (1), given $\varepsilon > 0 \exists R > 0 : |ye^{-y}| < \varepsilon$ for $y > R$.

Now, given $x \in (0, \delta)$, let $y = -\log(x)$. Then $y > -\log(\delta)$. Choose $\delta = e^{-R}$. Then $y > R$, which means that

$$|ye^{-y}| < \varepsilon \Rightarrow |x \log(x)| < \varepsilon.$$

Thus $\lim_{x \rightarrow 0^+} x \log(x) = 0$.

3. By definition, $x^x = e^{x \log(x)}$. Therefore,

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \log(x)} = e^{\lim_{x \rightarrow 0^+} x \log(x)} = e^0 = 1.$$

Definition 3.58: Landau Big-O

Given $f, g : D \rightarrow \mathbb{R}, x_0$ an accumulation point of D . We write $f(x) = O(g(x))$ as $x \rightarrow x_0$ if $\exists M > 0, \delta > 0$ such that

$$|f(x)| \leq M|g(x)|, \forall x \in D, |x - x_0| < \delta.$$

If $g \neq 0$, this means that $\frac{f(x)}{g(x)}$ is bounded near x_0 .

We write $f(x) = O(g(x))$ as $x \rightarrow +\infty$ if

$$\exists M > 0, R > 0 : |f(x)| \leq M|g(x)|, \forall x \in D \cap (R, \infty).$$

Example 3.59:

Consider the following examples.

- $x^2 = O(x)$ as $x \rightarrow 0$. This is correct as for $\delta = 1, |x^2| \leq |x|$ for $|x| < \delta$.
- $x = O(x^2)$ as $x \rightarrow 0$. This is incorrect as $\frac{|x|}{|x^2|} = \frac{1}{|x|}$, which is unbounded near 0.
- $x + x^2 = O(x)$ as $x \rightarrow 0$. This is correct as $|x + x^2| \leq |x| + |x|^2 \leq 2|x|$ for $|x| < 1$.
- $\frac{3x^3}{x^3+3} = O(1)$ as $x \rightarrow \infty$. This is correct as $\left| \frac{3x^3}{x^3+3} \right| \leq 3$ for all x .
- $\frac{3x^3}{x^3+3} = O(\frac{1}{x})$ as $x \rightarrow \infty$. This is incorrect as

$$\left| \frac{3x^3}{x^3+3} \right| = \frac{3x^4}{x^3+3} = 3x - \frac{9x}{x^3+3}.$$

This diverges as $x \rightarrow \infty$.

There also exists a related notation called little-o notation.

Definition 3.60: Landau Little-o

Given $f, g : D \rightarrow \mathbb{R}, x_0$ an accumulation point of D . We write $f(x) = o(g(x))$ as $x \rightarrow x_0$ if $\forall \varepsilon > 0, \exists \delta > 0$ such that

$$|f(x)| \leq \varepsilon|g(x)|, \forall x \in D, |x - x_0| < \delta.$$

Example 3.61:

Consider the following examples

- $x = o(x^2)$ as $x \rightarrow \infty$. This is correct as $\frac{x}{x^2} \rightarrow 0$.
- $x^2 = o(x)$ as $x \rightarrow 0$. This is correct as $\frac{x^2}{x} = |x| \rightarrow 0$.
- $\frac{3x^3}{2x^2+x^{10}} = o(1)$ as $x \rightarrow 0$. This is correct as

$$\frac{3x^3}{2x^2+x^{10}} = \frac{3x}{2+x^8} \rightarrow 0.$$

Example 3.62:

Reduce the following limit

$$\frac{x^3 - 7x^2 + 6x + 2}{x^2} = x - 7 + \frac{6}{x} + \frac{2}{x^2}.$$

Say you are interested in the limit as $x \rightarrow \infty$. It might be annoying to keep writing $\frac{6}{x} + \frac{2}{x^2}$, if we carry them through two pages of calculations. Instead, we can write

$$x - 7 + o(1) \text{ as } x \rightarrow \infty.$$

Or even if we consider for some reason only terms that grow we could write

$$x + O(1) \text{ as } x \rightarrow \infty = x + o(x) \text{ as } x \rightarrow \infty.$$

Example 3.63:

Given $f(x) = x + x^3 + 4x^4 + x^7, g(x) = x + \frac{3x^2}{1+x}$. We want to understand the behaviour of $f \cdot g$ near 0.

Solution. Observe that $f(x) = x + o(x^2)$ and $g(x) = x + O(x^2)$. Now $f \cdot g = (x + o(x^2))(x + O(x^2)) = x^2 + x \cdot O(x^2) + x \cdot o(x^2) + o(x^2) \cdot O(x^2)$.

Notice that $\frac{x \cdot o(x^2)}{x \cdot x^2} = 0$ implies that $x \cdot o(x^2) = o(x^3)$. Similarly, $x \cdot O(x^2) = O(x^3)$ and thus

$$f \cdot g = x^2 + O(x^3) + o(x^3) + o(x^4) = x^2 + O(x^3).$$

3.6 Sequences of Functions

Lec 17

We like to combine the ideas of sequences and functions.

Definition 3.64: Sequence of Functions

$D \subseteq \mathbb{R}$. $(f_n)_{n \geq 0}$ is a family of functions, $f_n : D \rightarrow \mathbb{R}$, indexed by $\mathbb{N}$.

To talk about limits of sequences of functions, we first need a notion of convergence of such sequences.

Definition 3.65: Pointwise Convergence

We say that $(f_n)_{n \geq 0}$ converges **POINTWISE** to $f : D \rightarrow \mathbb{R}$, if

$$\forall x \in D, f_n(x) \xrightarrow{n \rightarrow \infty} f(x).$$

Example 3.66:

Consider $f_n(x) = x^n$ on $D = [0, 1], n \geq 1$. Then if we fix $x \in [0, 1)$, then $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$. If we fix $x = 1$, then $f_n(1) = 1$ for all n . Therefore,

$$\lim_{n \rightarrow \infty} x^n = \begin{cases} 0 & x \in [0, 1) \\ 1 & x = 1. \end{cases}$$

So we see that the limit of continuous functions need not be continuous.

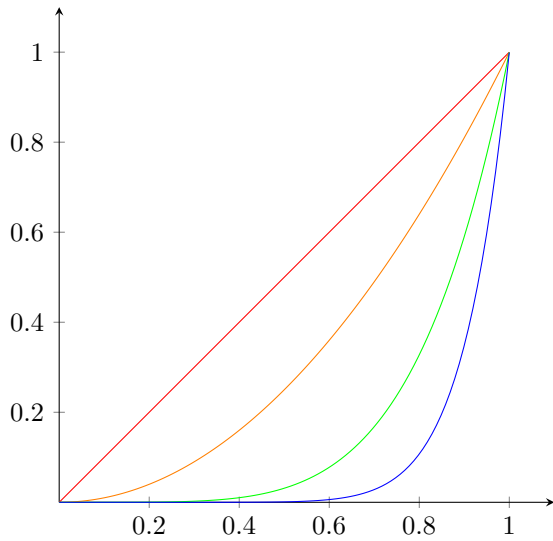


Figure 18: Family of functions

We can write pointwise convergence in terms of ε and δ .

$$\forall x \in D \forall \varepsilon > 0, \exists N \in \mathbb{N} : |f_n(x) - f(x)| < \varepsilon, \forall n \geq N.$$

However, note that N depends on x here. We thus define the following.

Definition 3.67: Uniform Convergence

f_n converges **UNIFORMLY** to $f : D \rightarrow \mathbb{R}$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : |f_n(x) - f(x)| < \varepsilon, \forall x \in D, \forall n \geq N.$$

The difference to pointwise convergence is that N does not depend on x .

Geometrically, this means that after some index N , all functions f_n lie within a band of width 2ε around f .

Theorem 3.68:

Given $f_n : D \rightarrow \mathbb{R}, f : D \rightarrow \mathbb{R}$. If f_n converges uniformly to f and all f_n are continuous, then f is continuous.

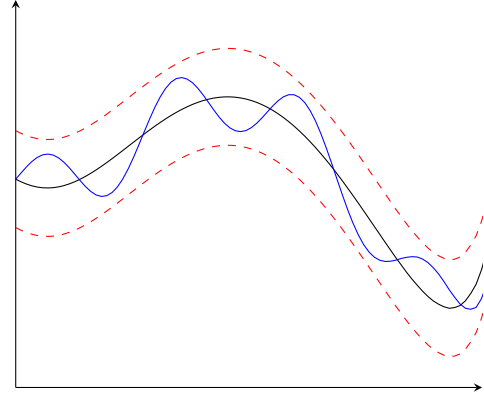


Figure 19: Uniform Convergence

Proof. Fix $\bar{x} \in D$, fix $\varepsilon > 0$. By uniform convergence, $\exists N$ such that

$$|f_N(y) - f(y)| < \frac{\varepsilon}{3}, \forall y \in D.$$

Since f_N is continuous, $\exists \delta > 0$ such that

$$x \in D, |x - \bar{x}| < \delta \Rightarrow |f_N(x) - f_N(\bar{x})| < \frac{\varepsilon}{3}.$$

Then,

$$\begin{aligned} |f(x) - f(\bar{x})| &= |f(x) - f_N(x) + f_N(x) - f_N(\bar{x}) + f_N(\bar{x}) - f(\bar{x})| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(\bar{x})| + |f_N(\bar{x}) - f(\bar{x})| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

□

Assume f_n converges pointwise to f . Then, we know

$$f(\bar{x}) = \lim_{n \rightarrow \infty} f_n(x).$$

Because f_n is continuous, we can write

$$f(\bar{x}) = \lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \bar{x}} f_n(x) \right).$$

On the other hand, we have

$$\lim_{x \rightarrow \bar{x}} f(x) = \lim_{x \rightarrow \bar{x}} \left(\lim_{n \rightarrow \infty} f_n(x) \right).$$

Therefore $f(\bar{x}) = \lim_{x \rightarrow \bar{x}} f(x)$ if and only if

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \bar{x}} f_n(x) \right) = \lim_{x \rightarrow \bar{x}} \left(\lim_{n \rightarrow \infty} f_n(x) \right).$$

4 Series and Power Series

Given a sequence $(a_n)_{n \geq 0}$, given $A \in \mathbb{R}$, we say that

$$\sum_{n=0}^{\infty} a_n = A \text{ if } s_n = \sum_{k=0}^n a_k \text{ converges to } A.$$

So $(s_n)_{n \geq 0}$ is a sequence and we look at its convergence. In this situation a_n is the n -th **TERM** of the series and A is the **SUM** of the series.

If the series has no limit, we say it does **NOT CONVERGE**. We say, that the series **DIVERGES** to $+\infty$ if $s_n \rightarrow +\infty$ and diverges to $-\infty$ if $s_n \rightarrow -\infty$.

For the moment being, we only want to consider $a_n \in \mathbb{R}$.

Proposition 4.1:

If $\sum_{n=0}^{\infty} a_n$ converges, then $a_n \rightarrow 0$ as $n \rightarrow \infty$.

Proof. Note that $s_n = s_{n-1} + a_n$. Thus $a_n = s_n - s_{n-1}$. Since $s_n \rightarrow A$, it follows that $a_n \rightarrow A - A = 0$. $\square$

Example 4.2: Geometric Series

Given $q \in \mathbb{R}$, consider the series (We let $q^0 = 1$)

$$\sum_{n=0}^{\infty} q^n.$$

Determine for which q this series converges and find the sum in that case.

Solution. If $|q| \geq 1 \Rightarrow |q|^n \geq 1 \forall n$, thus $q^n \not\rightarrow 0$. Therefore, the series does not converge.

If $|q| < 1$, look at $s_n = \sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$. Since $|q| < 1$, $q^{n+1} \rightarrow 0$. Thus, $s_n \rightarrow \frac{1}{1-q}$.

Example 4.3: Harmonic Series

Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges.

Solution. For $l \in \mathbb{N}$, look at the partial sum

$$\sum_{n=1}^{2^l} \frac{1}{n} = 1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq \frac{1}{2}} + \cdots + \underbrace{\frac{1}{2^{l-1}+1} + \cdots + \frac{1}{2^l}}_{\geq \frac{1}{2}} \geq 1 + \frac{l}{2}.$$

Thus, the partial sums are unbounded and the series diverges.

Lemma 4.4:

The sum $\sum_{n=0}^{\infty} a_n$ is convergent if and only if

$$\sum_{n=N}^{\infty} a_n.$$

is convergent for some $N \in \mathbb{N}$.

Proof. Fix N , call s_n the partial sums for $\sum_{n=0}^{\infty} a_n$ and $\tilde{s}_n$ the partial sums for $\sum_{n=N}^{\infty} a_n$. Then,

$$s_n = \sum_{k=0}^n a_k = \sum_{k=0}^{N-1} a_k + \sum_{k=N}^n a_k = \sum_{k=0}^{N-1} a_k + \tilde{s}_n.$$

$\square$

Proposition 4.5:

Let $a_n \geq 0 \forall n$. Then the sequence $s_n = \sum_{k=0}^n a_k$ is increasing. Therefore if s_n is bounded, then it converges. If s_n is not bounded, then it diverges to $+\infty$.

If $a_n \geq 0$, then s_n is bounded iff $\exists$ subsequence s.t. $(s_{n_k})_{k \geq 0}$ is bounded.

Lec 18

Proposition 4.6: Majorant and Minorant Criterion

Let $0 \leq a_k \leq b_k \forall k \in \mathbb{N}$. Then,

$$0 \leq \sum_{k=0}^{\infty} a_k \leq \sum_{k=0}^{\infty} b_k.$$

So if $\sum b_k$ converges, then $\sum a_k$ converges. Similarly, if $\sum a_k$ diverges to $+\infty$, then $\sum b_k$ does too.

Proof. $a_k \leq b_k \Rightarrow \sum_{k=0}^n a_k \leq \sum_{k=0}^n b_k \forall n$. By limit computation rules, the result follows. $\square$

In the above situation, we say that $\sum a_k$ is a **MINORANT** of $\sum b_k$ and that $\sum b_k$ is a **MAJORANT** of $\sum a_k$.

Remark 4.7:

Note that the Majorant and Minorant Criterion also works if $\exists N \in \mathbb{N}$ such that $0 \leq a_k \leq b_k \forall k \geq N$.

Example 4.8:

Consider $\sum_{k=1}^{\infty} \frac{1}{k^2}$. Observe that, for $k \geq 2$, $\frac{1}{k^2} \leq \frac{1}{k(k-1)}$. Then

$$\begin{aligned} \sum_{k=2}^{\infty} \frac{1}{k^2} &\leq \sum_{k=2}^{\infty} \frac{1}{k(k-1)} = \sum_{k=2}^{\infty} \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ &= \sum_{k=2}^{\infty} \frac{1}{k-1} - \sum_{k=2}^{\infty} \frac{1}{k} \\ &= \left(1 + \frac{1}{2} + \frac{1}{3} + \dots \right) - \left(\frac{1}{2} + \frac{1}{3} + \dots \right) = 1. \end{aligned}$$

Thus, by the majorant criterion, $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is bounded and therefore convergent.

Example 4.9:

Consider $\sum_{n=0}^{\infty} \frac{2n-10}{n^3-10n+100}$. For $n \geq N$, we have that $0 \leq a_n$ furthermore,

$$\begin{aligned} n^2 \frac{2n-10}{n^3-10n+100} &= \frac{n^3(2-\frac{10}{n})}{n^3(1-\frac{10}{n^2}+\frac{100}{n^3})} \\ &\rightarrow \frac{2-0}{1-0+0} = 2. \end{aligned}$$

So for n large enough, $n^2 a_n \leq 3$. Thus, $0 \leq a_n \leq \frac{3}{n^2}$. By the majorant criterion, $\sum a_n$ converges.

In exercise 4.8, we showed that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges. But we only could do this by doing this special trick that

$$\frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}.$$

But what if we had $\sum_{k=1}^{\infty} \frac{1}{k^{\frac{3}{2}}}$? We cannot do the same trick here.

Proposition 4.10: Cauchy Condensation Test

Let $a_k \geq 0$ and a_k be decreasing. Then

$$\sum_{k=0}^{\infty} a_k \text{ converges} \Leftrightarrow \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges}.$$

Proof.

$$a_0 + a_1 + \underbrace{\overbrace{a_2}^{\leq a_1}}_{\geq a_2} + \underbrace{\overbrace{a_3 + a_4}^{2a_2}}_{\geq 2a_4} + \underbrace{\overbrace{a_5 + \dots + a_8}^{\leq 4a_4}}_{\geq 4a_8} + \underbrace{\overbrace{a_9 + \dots + a_{16}}^{8a_8}}_{\leq 8a_{16}}$$

$$\sum_{k=0}^n 2^k a_{2^k} \geq \sum_{j=2}^{2^{n+1}} a_j \geq \frac{1}{2} \sum_{k=1}^{n+1} 2^k a_{2^k}.$$

In the above, the first term is the part at the top, the second term is the middle part and the last term is the bottom part. If we let $n \rightarrow \infty$, the result follows. $\square$

Example 4.11:

Consider $\sum_{k=1}^{\infty} \frac{1}{k^p}$. If $p \leq 0$, then $\frac{1}{k^p} \geq 1$ and thus the series diverges to $+\infty$. If $p > 0$, then $\sum \frac{1}{k^p}$ converges iff $\sum 2^k \frac{1}{(2^k)^p}$ converges.

$$\sum_{k=0}^{\infty} \frac{2^k}{2^{kp}} = \sum_{k=0}^{\infty} (2^{1-p})^k.$$

But this is a geometric series which converges iff $|2^{1-p}| < 1$, that is $p > 1$.

When we have series that can change sign, the following definition is useful.

Definition 4.12: Absolute Convergence

$\sum_{k=0}^{\infty} a_k$ converges **ABSOLUTELY** if

$$\sum_{k=0}^{\infty} |a_k| \text{ converges}.$$

We say that it converges **CONDITIONALLY** if

$$\sum_{k=0}^{\infty} a_k \text{ converges but } \sum_{k=0}^{\infty} |a_k| = +\infty.$$

Theorem 4.13: Riemann rearrangement theorem

Let $\sum a_k$ converge conditionally, fix $A \in \mathbb{R}$. Then $\exists \varphi : \mathbb{N} \rightarrow \mathbb{N}$ bijective, such that

$$\sum_{k=0}^n a_{\varphi(k)} \xrightarrow{n \rightarrow \infty} A.$$

So in essence, a conditionally convergent series can be re-arranged to converge to anything we want. The proof is considered extra material.

The important takeaway from this theorem is that when dealing with series, we should be very careful when re-arranging terms.

Definition 4.14:

Given $a_k \geq 0$, we define the **ALTERNATING SERIES** as

$$\sum_{k=0}^{\infty} (-1)^k a_k.$$

Proposition 4.15: Leibniz Criterion

If $a_k \geq 0$ and decreasing to 0, then the alternating series converges and

$$\sum_{k=0}^{2n+1} (-1)^k a_k \leq \sum_{k=0}^{\infty} (-1)^k a_k \leq \sum_{k=0}^{2n} (-1)^k a_k \quad \forall n \in \mathbb{N}.$$

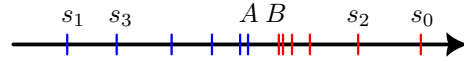


Figure 20: Leibniz Criterion

Proof. Let $s_n = \sum_{k=0}^n (-1)^k a_k$ be the partial sums.

For even numbers, we have

$$s_{2n+2} = s_{2n} - a_{2n+1} + a_{2n+2} \leq s_{2n}.$$

For odd numbers, we have

$$s_{2n+1} = s_{2n-1} + a_{2n} - a_{2n+1} \geq s_{2n-1}.$$

So the sequence $(s_{2n})_{n \geq 0}$ is decreasing and converging to some number B . Similarly, $(s_{2n+1})_{n \geq 0}$ is increasing and converging to some number A . We now show that $A = B$.

But we have $0 \leq B - A \leq s_{2n} - s_{2n-1} = a_{2n} \rightarrow 0$. Thus $A = B$ and the series converges. $\square$

Example 4.16: Alternating Harmonic Series

Consider $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$. By the Leibniz criterion, this series converges. However, $\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k}$ diverges. Thus, the alternating harmonic series converges conditionally.

Proposition 4.17: Cauchy Criterion

A series $\sum a_k$ converges iff $\forall \varepsilon > 0 \exists N$ such that

$$\left| \sum_{k=m+1}^n a_k \right| < \varepsilon \quad \forall n > m \geq N.$$

Proof. $\sum_{k=0}^{\infty} a_k$ converges iff $s_n = \sum_{k=0}^n a_k$ converges. This however is equivalent to $(s_n)_{n \geq 0}$ being a Cauchy sequence, i.e.

$$\forall \varepsilon > 0 \exists N : |s_n - s_m| < \varepsilon \quad \forall n > m \geq N.$$

However, $s_n - s_m = \sum_{k=m+1}^n a_k$. Thus the result follows. $\square$

4.1 Absolute Convergence

We now want to study absolutely converging series a bit more.

Proposition 4.18:

If $\sum a_k$ converges absolutely, then it converges and

$$\left| \sum_{k=0}^{\infty} a_k \right| \leq \sum_{k=0}^{\infty} |a_k|.$$

Proof. Since $\sum_{k=0}^{\infty} |a_k|$ converges, $\forall \varepsilon > 0 \exists N$ such that for all $n \geq m \geq N$, $\sum_{k=m+1}^n |a_k| < \varepsilon$.

By triangle inequality,

$$\left| \sum_{k=m+1}^n a_k \right| \leq \sum_{k=m+1}^n |a_k| < \varepsilon \quad \forall n \geq m \geq N.$$

So by Cauchy, $\sum_{k=0}^{\infty} a_k$ converges.

Also $|s_n| = \left| \sum_{k=0}^n a_k \right| \leq \sum_{k=0}^n |a_k|$. Letting $n \rightarrow \infty$, the result follows. $\square$

Lec 19

Remark 4.19:

If $(x_n)_{n \geq 0}$ is a sequence converging to $\alpha \in \mathbb{R}$, then:

- $\forall q > \alpha, \exists N : x_n < q \quad \forall n \geq N$
- $\forall r < \alpha, \exists N : x_n > r \quad \forall n \geq N$

The following proposition is inspired by trying to compare a series with a geometric series.

Proposition 4.20: Cauchy Root Criterion

Given $(a_n)_{n \geq 0}$, we want to understand the absolute convergence of the series $\sum_{k=0}^{\infty} a_k$. Define

$$\alpha := \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

Then, if $\alpha < 1$, the series converges absolutely. If $\alpha > 1$, the series does NOT converge.

Proof. First, assume $\alpha < 1$. Define $q = \frac{1+\alpha}{2}$. So $q \in (\alpha, 1)$. By definition of lim sup,

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \left(\sup_{k \geq n} \sqrt[k]{|a_k|} \right).$$

By the above remark, $\exists N$ such that

$$\sup_{k \geq n} \sqrt[k]{|a_k|} < q \quad \forall n \geq N.$$

But this means that $\sqrt[k]{|a_k|} < q$ for all $k \geq N$, which is the same as $|a_k| < q^k$ for all $k \geq N$. Since $q < 1$, $\sum_{k=0}^{\infty} q^k$ converges. By the majorant criterion, $\sum_{k=0}^{\infty} |a_k|$ converges absolutely.

Now, assume $\alpha > 1$. Recall that the lim sup is an accumulation point. Thus, $\exists$ subsequence $(\sqrt[n_k]{|a_{n_k}|})_{k \geq 0}$ such that

$$\sqrt[n_k]{|a_{n_k}|} \rightarrow \alpha > 1.$$

By the above remark with $r = 1$, $\exists K$ such that

$$\sqrt[n_k]{|a_{n_k}|} > 1 \quad \forall k \geq K.$$

But this is equivalent to $|a_{n_k}| > 1$ for all $k \geq K$. Thus, the numbers a_n do not go to 0 and thus the series cannot converge. $\square$

Remark 4.21:

Consider the series $\sum_{k=1}^{\infty} \frac{1}{k}$. This series has $\alpha = 1$, but diverges.

The series $\sum_{k=1}^{\infty} \frac{1}{k^2}$ has $\alpha = 1$ but converges.

So for $\alpha = 1$, no conclusion can be drawn.

Proposition 4.22: D'Alemberts Quotient Criterion

Assume $a_0 \neq 0 \forall n$. Let

$$\alpha = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

Then $\alpha < 1$ implies absolute convergence, $\alpha > 1$ implies no convergence.

Proof. Assume first, $\alpha < 1$. Let $q = \frac{1+\alpha}{2} \in (\alpha, 1)$. Then, by the remark, $\exists N$ such that

$$\left| \frac{a_{n+1}}{a_n} \right| < q \quad \forall n \geq N.$$

Then $|a_n| = \frac{|a_n|}{|a_{n-1}|} \cdot \frac{|a_{n-1}|}{|a_{n-2}|} \cdots \frac{|a_{N+1}|}{|a_N|} |a_N| < q^{n-N} |a_N|$ for all $n \geq N$.

But all of these terms must be $< q$. So for $n \geq N$,

$$|a_n| < q^{n-N} |a_N| = \frac{|a_N|}{q^N} q^n.$$

So the series $\sum_{n \geq N} |a_n|$ is bounded by $\frac{|a_N|}{q^N} \sum_{n \geq N} q^n$ which converges since $q < 1$. Thus, by the majorant criterion, $\sum |a_n|$ converges.

The opposite case, $\alpha > 1$ implies by the remark with $r = 1$ that

$$\left| \frac{a_{n+1}}{a_n} \right| > 1 \quad \forall n \geq N.$$

So $|a_{n+1}| > |a_n|$. But this implies in particular that $|a_n| > |a_{n-1}| > \cdots > |a_N| > 0$ for all $n \geq N$. Thus, the terms do not go to 0 and thus the series cannot converge. $\square$

Again for $\alpha = 1$, no conclusion can be drawn.

Proposition 4.23:

Let $\sum_{k=0}^{\infty} a_k$ be absolutely convergent. Let $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be bijective. Then $\sum_{k=0}^{\infty} a_{\varphi(k)}$ is absolutely convergent and

$$\sum_{k=0}^{\infty} a_{\varphi(k)} = \sum_{k=0}^{\infty} a_k.$$

Proof. Fix $\varepsilon > 0$. Since $\sum |a_k|$ converges, $\exists N$ such that

$$\sum_{k=N+1}^{\infty} |a_k| < \frac{\varepsilon}{2}.$$

Let M be the smallest number s.t. $\{0, \dots, N\} \subseteq \{\varphi(0), \dots, \varphi(M)\}$. Equivalently, $M = \max\{\varphi^{-1}(0), \dots, \varphi^{-1}(N)\}$. Then,

$$\sum_{k=0}^M a_{\varphi(k)} - \sum_{k=0}^N a_k = \sum_{\substack{\varphi(k) > N \\ 0 \leq k \leq M}} a_{\varphi(k)}.$$

Therefore, for $n \geq M$,

$$\begin{aligned}
\left| \sum_{l=0}^n a_{\varphi(l)} - \sum_{k=0}^{\infty} a_k \right| &= \left| \sum_{l=0}^n a_{\varphi(l)} - \sum_{k=0}^N a_k - \sum_{k=N+1}^{\infty} a_k \right| \\
&\leq \left| \sum_{\substack{\varphi(l) > N \\ 0 \leq l \leq n}} a_{\varphi(l)} \right| + \left| \sum_{k=N+1}^{\infty} a_k \right| \\
&\leq \sum_{\substack{\varphi(l) > N \\ 0 \leq l \leq n}} |a_{\varphi(l)}| + \sum_{k=N+1}^{\infty} |a_k| \\
&\leq 2 \sum_{k=N+1}^{\infty} |a_k| < \varepsilon.
\end{aligned}$$

So the sum $\sum_{l=0}^n a_{\varphi(l)}$ converges to $\sum_{k=0}^{\infty} a_k$ as $n \rightarrow \infty$. Applying the same to $\sum |a_k|$ shows absolute convergence. $\square$

An interesting question is how we can compute the product of two series.

Theorem 4.24:

Let $\sum a_k$ and $\sum b_k$ be absolutely convergent series. Fix $\alpha : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ bijection. Then

$$\sum_{k=0}^{\infty} a_k \cdot \sum_{k=0}^{\infty} b_k = \sum_{k=0}^{\infty} a_{\alpha_1(k)} b_{\alpha_2(k)}.$$

and the product series converges absolutely.

Proof. Consider first the following bijection:

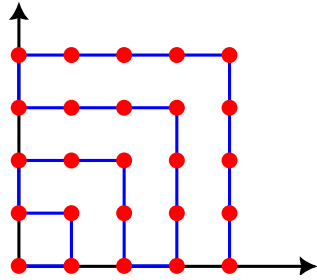


Figure 21: Specific Bijection α

Then,

$$\begin{aligned}
\sum_{k=0}^{n^2-1} |a_{\alpha_1(k)}| |b_{\alpha_2(k)}| &= \sum_{k=0}^{n-1} |a_k| \sum_{k=0}^{n-1} |b_k| \\
&\leq \left(\sum_{k=0}^{\infty} |a_k| \right) \left(\sum_{k=0}^{\infty} |b_k| \right) < \infty. \\
\Rightarrow \sum_{k=0}^{\infty} |a_{\alpha_1(k)}| |b_{\alpha_2(k)}| &\leq \infty.
\end{aligned}$$

So the series $\sum_{k=0}^{\infty} |a_{\alpha_1(k)} b_{\alpha_2(k)}|$ converges absolutely which implies that it converges.

Again, note that $\sum_{k=0}^{n^2-1} a_{\alpha_1(k)} b_{\alpha_2(k)} = \left(\sum_{k=0}^{n-1} a_k \right) \left(\sum_{k=0}^{n-1} b_k \right)$. By definition, both limits on the right converge by assumption. The limit on left we have just shown to exist. Thus, both limits are equal for this specific bijection.

But by proposition 4.23, the result holds for any bijection. $\square$

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